FINDIEMME
ADHESIVE TECHNOLOGIES

User Manual

Dynamix

Revision: 1.0 Date: 24/12/2016

Index by Section

0. Identification of the machine

Located on the machinery you will find an Identification plate which includes the machine identity description along with power connection details and weight.

Example Below:

IMPORTANT WARNING

THE OBSERVANCE OF THE INSTRUCTIONS REPORTED IN THE HANDBOOK ALLOWS THE OPERATOR TO PERFORM HIS WORK WITHOUT ANY PROBLEMS AND AVOIDING THE RISK OF ACCIDENTS.

FAILURE TO OBSERVE THESE INSTRUCTIONS CAN RESULT IN VERY SERIOUS HARM TO THE OPERATOR.

WE REMIND YOU THAT THE ELECTRIC SYSTEM OF THE MACHINE IS NOT FLAME PROOF, SELECTION OF CLEANING PRODUCTS AND ADHESIVES SHOULD CONSIDER THE RISK OF FIRE AND OPERATOR SAFETY.

TAKE CARE OF ALL MOVING COMPONENTS.

1. Technical Specification

Dynamix 240	1 x 200Ltr Drum unloader (Piston Type) 2 x 20Ltr Drum unloader (Piston/Membrane/Tank)
Dynamix 440	2 x 200Ltr Drum unloader (Piston Type) 2 x 20Ltr Drum unloader (Piston/Membrane/Tank)
Dosing System **Gear Pump:** **Reducer:**	 3cc / 7cc / 10cc 10:1 / 28:1 / 40:1
Mixing Method	Static Mixer MS10-24 / MS13-24
Application Head Type	Precision Ezi-Mix Ezi Mix Pro Snuff-Bak
Monitoring	Pressure Sensor Mass Flow meter
Power Supply	3Phase + Neutral + Earth. 380-415VAC 16 AMP
Pneumatic	6-8 Bar Dry air

2. General Machine Description

The adhesive application equipment has been designed as a retro fit system to either replace existing adhesive application equipment or as a module to be integrated into new build lines produced by alternative suppliers / system integrators.

Fin Diemme is responsible for the functionality of the machinery supplied and cannot be held responsible for any issues arising from other none supplied machinery involved in the line.

The equipment consists of 4 modules:

- **A Component Transfer System.** Depending the configuration opted for, this may consist of one or two PF200 transfer pump systems designed for the transfer of medium to high viscosity Resin from 205Ltr packaging to a precision dosing system. The PF200 utilises a piston pump mounted onto a follower plate that is raised and lowered into the drum using pneumatic pistons.

- **B Component Transfer System.** Depending on the adhesive technology being used and the customer preferences this may consist of a variety of configurations including the following options:

  2 x PF20M Two x 20Ltr Pail unloaders with Membrane transfer pump technology and a dip pipe mounted on a piston for drum changes. Typically used for moisture sensitive medium viscosity hardeners such as filled MDI and Polyurethane Pre-polymers.

  2 x PF20P Two x 20Ltr Pail unloaders with Piston transfer pump technology mounted on a follower plate with that is raised and lowered into the drum using pneumatic pistons. Typically used for less sensitive higher viscosity hardeners such as some Silicones.

  1 x PF20M + 1 x V70M 1 x 20Ltr Pail unloader with Membrane transfer pump technology and a dip pipe mounted on a piston for drum changes feeding a tank system with membrane pump to feed the dosing system. Typically used for moisture sensitive medium viscosity hardeners such as filled MDI and Polyurethane Pre-polymers.

- **Dosing System.** This consists of a precision gear pumping system for A and B components the configuration of which can be adjusted to optimize accuracy based on the pump volume required.

- **Application Head.** The adhesive is dosed to the applicator head which controls the stop/start of the dosing and the mixing of the two components. A range of applicator heads are available with differing features to suit the customer preference, for more details on the most suitable gun selection contact your Fin-Diemme Technician.

3. Installation

Installers must be skilled personnel who have been specifically trained to accomplish the required operations with competence and compliance with safety requirements. It is recommended that all installation is carried out under the supervision of the manufacturer or by the manufacturers own personnel.

It is recommended to leave a clearance of approx. 1.5m around the machine for access.
At the installation the floor must be smooth and free of holes or depressions.
It is extremely important that the installation site is adequately lit and ventilated.

Conditions for Use – Environmental Conditions

The machine should be used in rooms with a temperature range of 15 - 35 DegC and a relative humidity lower than 20 @ 65 %.
The machine must be used in a place protected from external weather conditions.

In the place of installation lighting must be appropriate, in order to allow the correct use and maintenance of the machine. The user shall be responsible for complying with the local legislation in force regarding lighting.

It is strictly forbidden to use the machine in explosive or partially explosive environments or in environments where there are flammable gases or liquids.

Verify that the terminal connectors are tight as during transport vibrations could slacken or disconnect the cables.

Before starting the machine, check the exact connection of the motors to the mains voltage:

- The power supply must be appropriately rated to sustain the maximum electrical input of the machine. It shall not be lower than the value marked in the machine's rating plate and indicated and in this manual..
- Check that the characteristics of the supply line are compatible with those of the machine, namely **415 Volt, 50Hz (three-phase + neutral + ground)**. During connection make sure that the neutral and ground wires are inserted in the relevant connecting terminals. Refer to the electrical wire diagrams of the machine to identify the feed clamps (connecting terminals).

Important

A wrong connection of the neutral wire may seriously damage the electronic parts of the machine.
Check to make sure that the power supply line is suitable for the operating and start-up load.

Grounding: The ground system must comply with the requirements of norm CEI 64-8.
The power cable section must be suitable for the rated capacity of the machine.

To open the electrical board, turn the machine's main switch to the O position. Use the special keys supplied with the machine to open the door. Make sure that the external supply to the machine is isolated before entering. Only qualified electricians with knowledge of the system should enter the panel.

Pneumatic Connection

Connect the air supplies. These should be dry and with a supply pressure of 6-8Bar.

Air volume should be sufficient to avoid starvation during any operation.

Connect the air feeding pipe coming from the compressor or the pneumatic distribution line to the rapid connector. Use a feeding pipe with an inside diameter not smaller than 8 mm.

Commissioning the machine

It is recommended that the commissioning be carried out by the manufacture of the machinery or a member or personnel trained on the machinery by the machine manufacturer.

Note: That the machinery is able to be used in a variety of configurations for which the procedures may vary a little. If you are unsure always contact Fin-Diemme prior to use. The following information is given as a general guideline for a typical line.

Pressing Emergency Stops

Immediately blocks the whole machine. It is necessary to release the emergency button and then press the “Reset” button on the main control panel before you will be able to resume production.

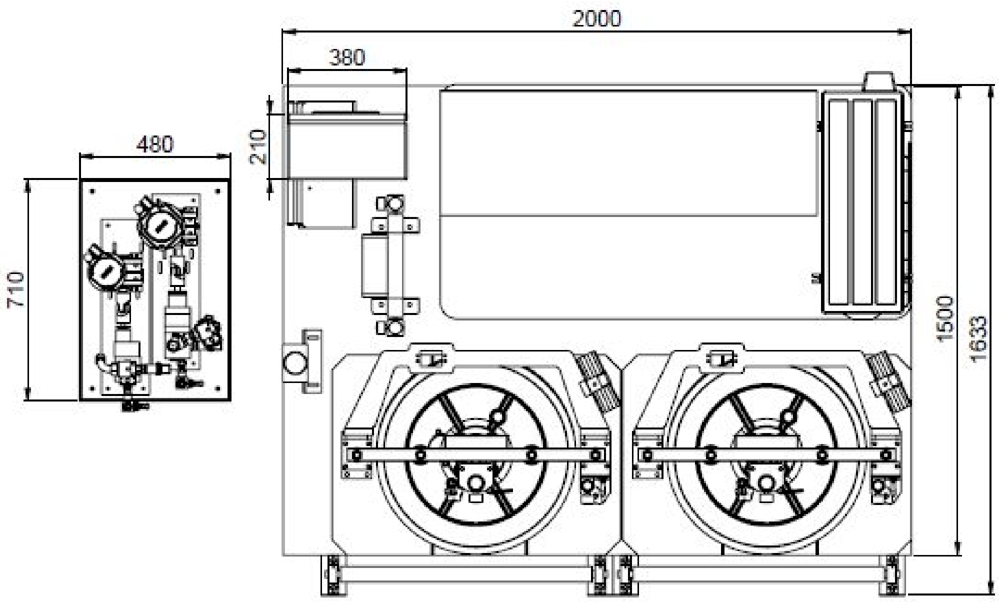
The following safety switches are available:
- Emergency stop button on the main control panel.
- External command. To loop the glue machinery into the existing line E. Stop or to add E-stops in other positions enter the loop as per the wiring diagram on plug “External Commands” Pin 1 and 2 removing the loop in the plug.

Faulty machinery

The machine has been designed and built so as to ensure a proper and safe operation over a long period of time.
In any case, for a longer life it is necessary to follow the indications regarding maintenance reported in this handbook.
It is also indispensable to periodically check all safety switches and zero switch sensors.

Access

Access for maintenance and loading of material is required on all four sides of the machine and a clearance of 90-100cm should be allowed around the perimeter of the equipment.



4. Maintenance and Controls

Adjustment / Calibration

IMPORTANT Gear Pumps

The sealing system for the gear pumps utilises a graphite impregnated cloth packing which during the early part of its life relaxes whilst the gear pump is turning and the packing is bedding in. It is therefore compulsory to tighten the nuts on the compression flange of the packing to ensure that a seal is maintained. This should be carried out as required however in the early part of the life of the pump it should be carried out daily.
To carry out this simply tighten the bolts on the compression flange. Tighten opposite bolts to ensure the compression is even. Take care not to over tighten them to a point where the motor load is too great.
The bolts can be seen in "Section 6 – 7cc/3cc pumps and is referring to the component 14 (3cc) and 17 (10cc) bolts.

Type		Tightening torques (N*m)					
Screw	Pitch (mm)	8,8		10,9		12,9	
		dry	with HTC*	dry	with HTC*	dry	with HTC*
M 4	0.7	3.1	2.4	3.9	3	4.1	3.2
M 5	0.8	6.1	4.8	7.7	6	8.1	6.4
M 6	1	11	8.3	13	10	14	11
M 8	1.25	26	20	32	25	34	27
M 10	1.5	51	40	53	50	68	53
M 12	1.75	88	69	110	86	115	91
M 14	2	140	110	175	135	185	145
M 16	2	220	170	275	210	290	225
M 18	2.5	305	235	380	295	400	310
M 20	2.5	430	330	540	415	570	440
M 24	3	740	570	920	720	980	760
M 30	3.5	1500	1150	1850	1450	1950	1500

* = Lubricant for high temperature (ex. Unimoly HTC Metallic-Paste, Klüber)

Applicator head

See “Section 6 - Applicator Head - Ezi Mix”. Component number 1 is made from a special plastic which is malleable to ensure a complete seal against the metal seat. This component is prone to wear and may require changing from 3-6 months of use.
It is important to regularly ensure that the components 2/3/4 and the seat 11 are kept free from contamination. In the event of starvation of either A or B component during start up, maintenance or extrusion there is starvation or either component, due to the back pressure caused from the static mixer ahead of the flow there us a risk that the material cross contaminates. The first sign of this is increases line pressure between the gear pump and applicator head. This should be check regularly as part of a preventative breakdown procedure.

Piston Pump Seals

The life expectancy of the seals on the piston pump (section 6.14 items 8 & 9) can vary greatly based on the material being pumped (abrasive fillers, chemical reactions), the pressure in the system (active and idle) and the other factors such as lubrication oil selection and sedimentation of adhesive. These may need to be replaced on a regular basis.

System Checks

Task	Daily	Weekly	Monthl y	3 Month
Static Mixer change	> 25 Deg C 30 minutes < 25 Deg C 45 minutes			
Mix ratio check	Minimum once per shift.			
Replace mixer nozzle with Mesamol cap.	When not in use.			
Check and record operating pressure during production, if there is a noticeable deviation from the standard extrusion pressure, contact experienced local maintenance or Fin Diemme for advice.	Recommended > twice per shift.			
Gear Pump Packing Seals (check for leakage)*		X		
Lubricate Piston Pump shaft with inert oil (e.g. Mesamol)		X		
Observe piston movement is smooth during operation.			X	
Check for any build up of cured B component on Dip pipe.	During drum changes.		X	
Check o ring seals on A component Platen for damage/leakage.	During drum changes.		X	
Check cooler water level.				X
Check cooler functioning and chilled cabinet temperature.	X			
Check 3 Way valve movement, Indicator should fully rotate.				X
Driver check (“Mode” – “OB” – “Up x 7”), collate value on start up, and monitor the AMP Max %.				6 Month
Check for general Air and pipe leakage on all joints.	X			
Check Supply pressures to all transfer pumps versus set up.		X		

5. Safety

The machine is provided with devices that assist in the safety of operators against accidents.
In particular, the following devices have been designed:

- EMERGENCY mushroom button on the push-button panel; if pressed it cuts out the power supply to the whole machine.

IMPORTANT
As the machine is being supplied as a component of a larger line, it is necessary for the customer to integrate safety guarding to encompass the supplied modules within the existing line to protect the operators from any moving components. Access ways can then be integrated either into the External emergency or ready circuits in the external command circuits to prevent any mechanical movement when safety guards are removed.
After the intervention of one of the above safety devices, press the "RESET" button (it must turn on) so as to set the rollers moving again.

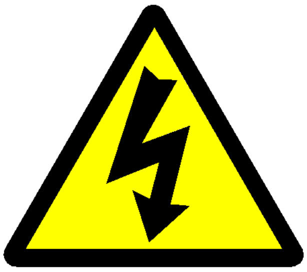
IMPORTANT
The machinery is designed for the application of adhesives. Many adhesive will require operator protection for protection against fumes, released toxic components or atomised particles.
Extraction of fumes and atomised particles is to be supplied by the customer not the machine supplier.
It is recommended that the customer liaise with the adhesive supplier to clearly understand the safety protection required.

Safety extraction should be in place prior to the installation to allow the machine commissioning in a safe manner should this be highlighted in the adhesive supplier 16 CHIP safety data sheet.

The machine has been designed with the utmost care to make it safe throughout production, transport, adjustment and maintenance operations. Nevertheless not all risks for either operators or the environment could be eliminated, be it for technological reasons (material reliability) or management-related issues (overcomplicated operations).

Therefore, when operating the machine the following residual risks - as well as those connected with its use - must be taken into due consideration:

Electrical Hazard - High Voltage Present 415V



General Hazard – Moving Components, Potential crushing hazard

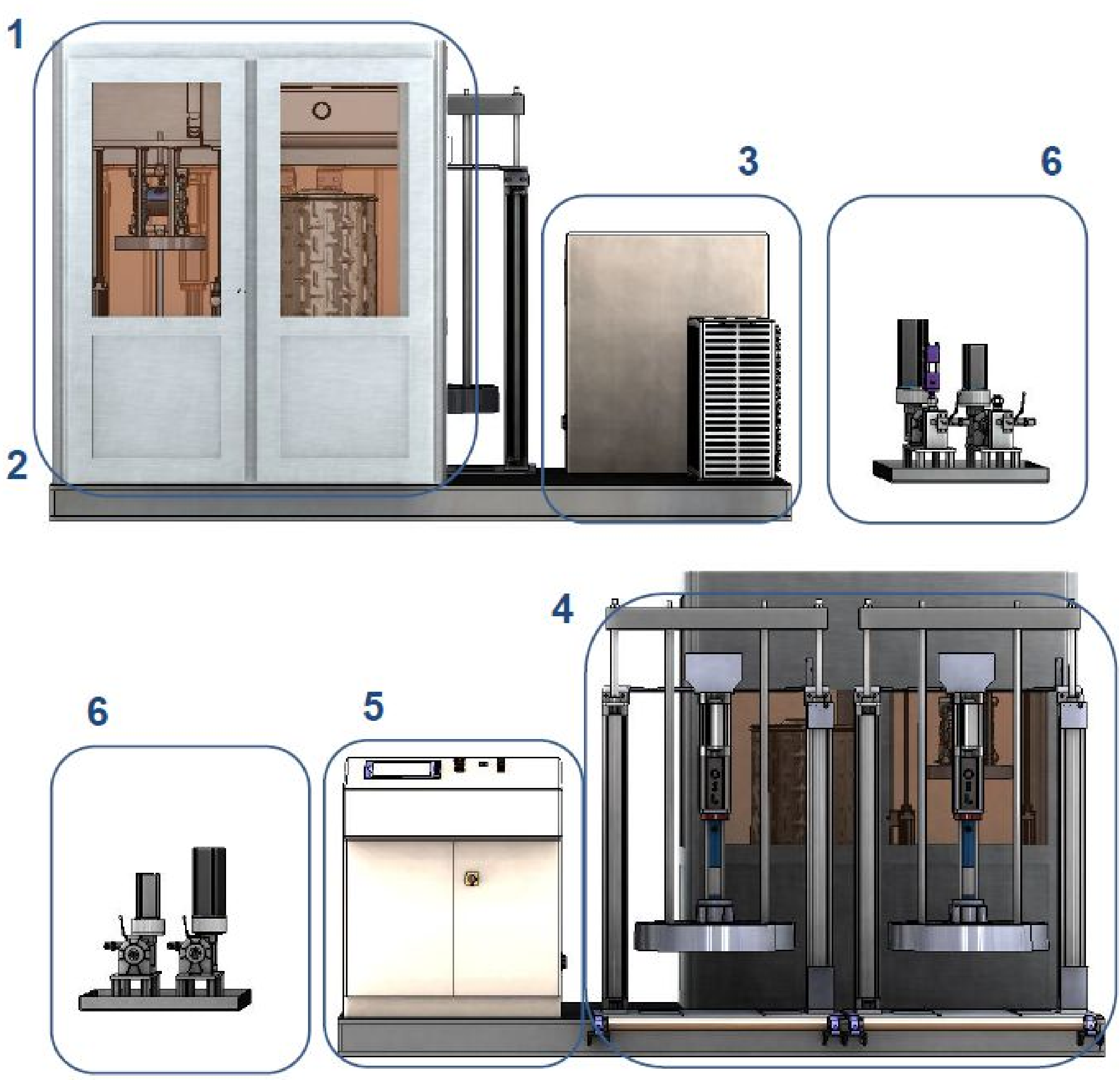
  

Contact with Chemicals – Refer to the adhesive suppliers COSH 16 Part Safety Data Sheet for required safety protection.

High Temperature – Potential hot surfaces

6. Machine Components

6.1 The basic configuration of all models is as follows:



1.	B Component Transfer Pump
2	Climate control Environment
3.	Air Conditioning Unit (in Compact Line this is mounted above the B Component cabinet)
4.	A Component Transfer Pump
5.	Operator Control Panel (in Compact Line this is mounted above the B Component cabinet)
6.	External Dosing Pump system (is fixed on the machine for transportation, however this is recommended to be positioned as close as possible to the application in the Glue cell area)
7.	Plus Dry air unit, heated hose, pneumatic cabinet, drum loading system……..

Connection Set Up

6.2 DMIX440 Compact – Amenities

Connect the amenities as follows:

Power

The unit requires a power supply 3 Phase + Neutral + Earth. The supply should be suitable for 16 Amps. The unit is supplied with a plug to suit these requirements which connects into the control panel.

Air

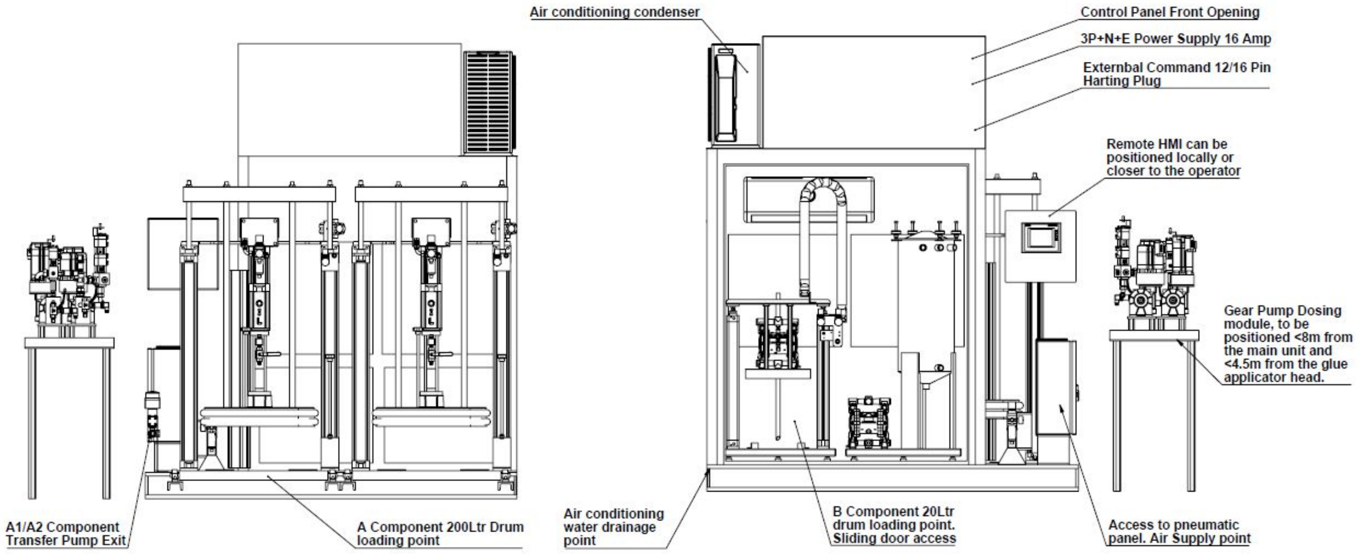
The unit has an enclosure for the pneumatic components fixed to the base of the machinery. On the external of the control panel there is a quick lock connector to fit standard PE 8mm diameter airline. This should be connected to a dry air supply with 6-8 Bar and substantial volume to avoid any pressure loss.

Drainage

When using an Air conditioning unit for the climate control cabinet of the adhesive it is necessary to connect the water outlet to a suitable drain or container to collect the water.

Communication

The labelled “External Command” 12 pin “Harting type” plug (supplied) on the side of the control panel provides basic communication with the controlling line where applicable. The signals are all clean contact and provide; machine ready, machine in automatic, machine alarm outputs and receive start A/B Component gear pumps/gun and an emergency stop loop, for further details refer to the electrical wiring diagram or contact Fin-Diemme.



6.3 Standard Start Up

DMIX440 Compact

This start up procedure is as follows:

Power On

Note: In Normal working practise where the adhesive is maintained in a climate controlled environment within the dosing equipment, once the adhesive is loaded in the system the equipment must remain switched on, with the exception of short term switching off for maintenance.

Start Up / Auxiliary inserted

When pressed this gives the power to the various electrical components inside the electrical panel.
If an emergency button is enabled, once the emergency has been cleared it will be necessary to press this again to restart.

If the PLC is booted prior to pressing “Start Up”, there will be an alarm on the PLC. First press the “Start Up” button and then clear the Alarm from the PLC by pressing the alarm icon on the screen.

Automatic (See “Software-Automatic”)

Enter the automatic ambient on the PLC. In this mode the equipment is ready to receive the signal from the Robot in the Glue cell. Pressure maintain (if selected) will be enabled in this ambient.

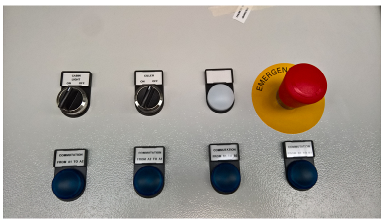
If the machine is being switched on for the first time:

Climate Control B Component

Switch on the power via the selector switch on the control panel and then fix the temperature mode via the Split remote controller. The setting mode is dependent upon the factory conditions. For warmer environments switch to standard Air conditioning and set the temperature to 20 DegC. In cold environments switch to heating and set the temperature at 20 DegC. If the factory conditions fluctuate take care to control the settings of Climate control zone.

Light

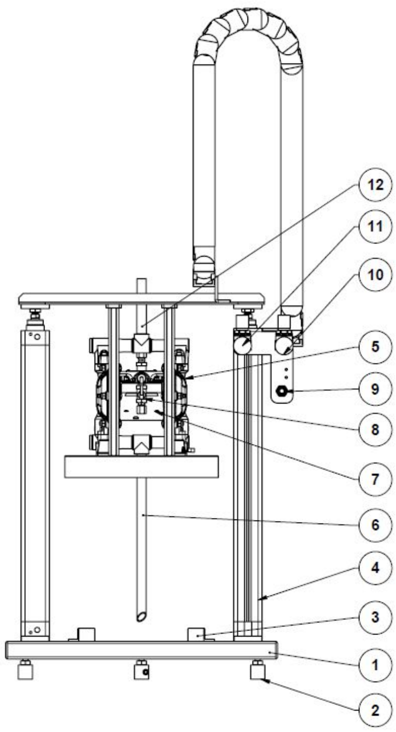
Switch on the power to the B Component cabinet light via the selector switch on the control panel.



Connection Set Up

6.4 DMIX440 Compact – B Component

Loading the adhesive in the system:

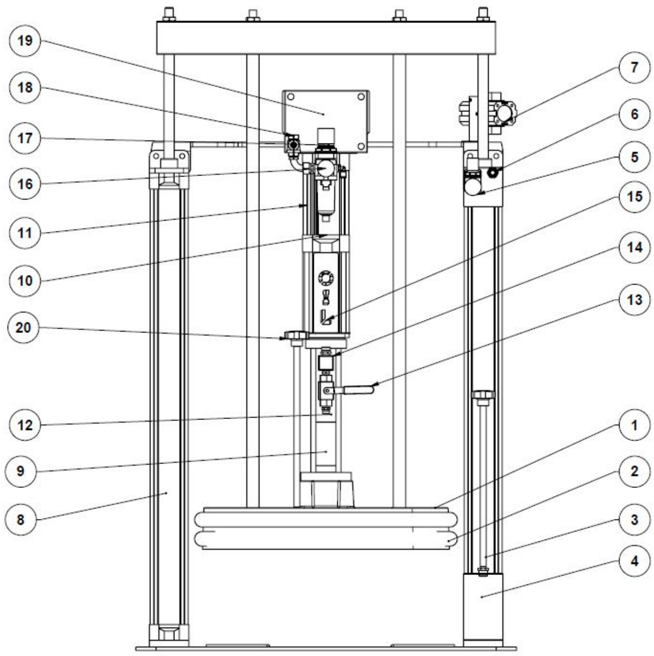


0. Open the sliding door to the left of the B component environmental controlled cabinet.
1. Lift the membrane pump (7) to the maximum height by placing the lever (9) in the upward position with the air pressure (11) set at 3-4 bar.
2. Remove the clip for the adhesive drum (glue supplier dependent) but leave the lid in place. Push the drum until you hit the positioning feet (3 these will be set up and locked in the correct position by the Fin-Diemme technician during installation).
3. Lower the membrane pump (7) onto the drum by placing the lever (9) in the downward position, it will pierce the lid and continue to travel to the maximum low position. If the lid is not pierced increase the pressure at the air regulator (11).
4. Increase the air pressure to the membrane pump (10) and open the purge valve (8) until a continuous flow of adhesive without any air pockets exits the purge valve. The close the purge valve and set the machine to automatic for a standard change or continue to purge throughout the system via the main PLC.

Connection Set Up

6.5 DMIX440 Compact – A Component

Loading the adhesive in the system:



1. If it's not the first time and the low level sensor has been enabled signifying the drum should be changed, first switch off the air supply to the piston pump (17/18).
2. Raise the platen out of the drum by switching the lever (6) into the upward position with the air pressure (5) set to maximum pressure (6-8Bar). If this is not the first time it is necessary to remove the wand (20) to allow air to enter into the drum and prevent the drum from collapsing due to a vacuum being created as the platen raises. If the platen still will still not raise then replace the removed wand with the wand (3) and open the valve found on the top of the wand for a short period of time (2-3 seconds) to allow air to be injected under the platen forcing the platen upwards and the empty drum downwards.
3. With the platen fully raised, remove the old drum (if necessary) and place the new drum with the lid complete removed under the platen. The positioning of this will be against the feet set up by the Fin-Diemme Technician.
4. Lower the platen with both wands removed into the drum by positioning the lever (6) in the downwards position and allow the air to evacuate. Monitor the exit of the platen (where the wand was removed) and when the adhesive can be seen to move and prior to it exiting the hole, replace the wand 20) tightly.
5. Switch on the air supply to the piston pump (17/18), remove the cap from the purge valve (12) and purge the material via the two valve (13) by moving it to the downwards position. When all the air has been evacuated from the system close the two way valve (13) by moving it to the horizontal position.

FINDIEMME ADHESIVE TECHNOLOGIES

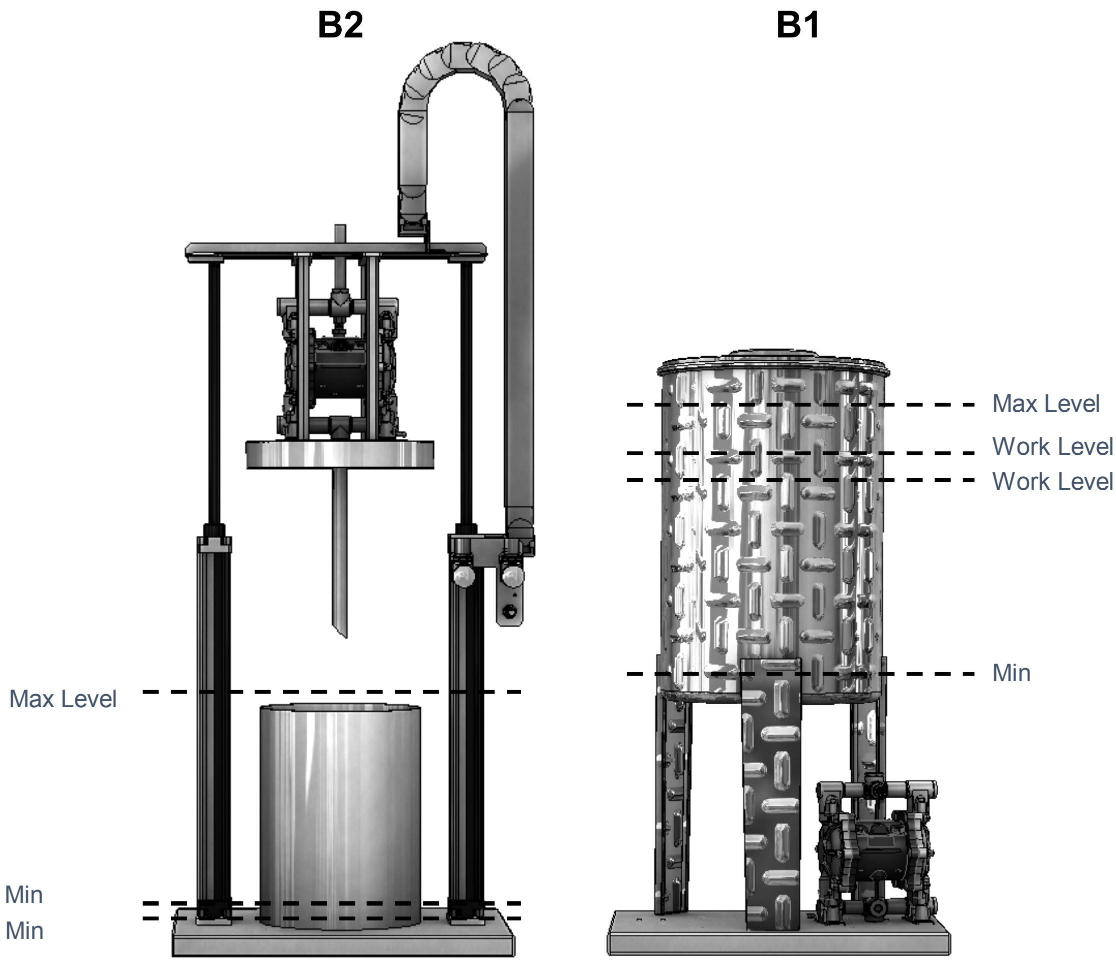
6. Connection Set Up

6.6 DMIX440 Compact – B Component Option 1 x PF20M + 1 x V70M

System Control Parameters:

The principle of this option is that the "B2" pumping system loads the "B1" tank/transfer pump system. By using the tank combined with the standard unloading system you avoid the possibility of air in the line during drum changes, you have the option for continuous running during drum changes and with the stirring option you have the chance to mix the product during use improving the homogenous viscosity of the product.

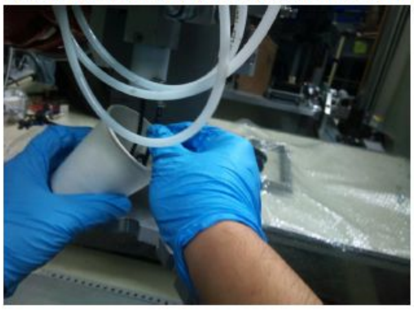
The parameters for setting up the unit can be found in "Setting Machine" – "Load Cells B1 and B2".



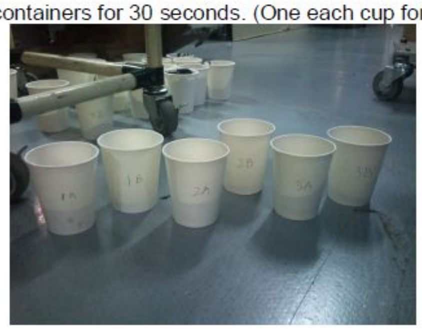
6.7 Connection Set Up

Mix Ratio Check method: See “Software-Weight test” for the best PLC ambient for carrying out ratio and repeat-ability testing.

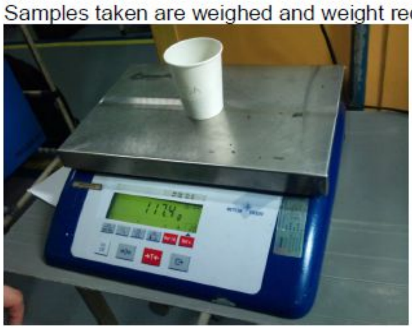
1) Product is purged for 10 seconds into a container. This is done to release the pressure built up in the nozzle when the valve is closed. This will help to reduce risk of product splash.



2) Product is dispensed into 2 individually marked containers for 30 seconds. (One each cup for Part A and B)



3) Samples taken are weighed and weight recorded.



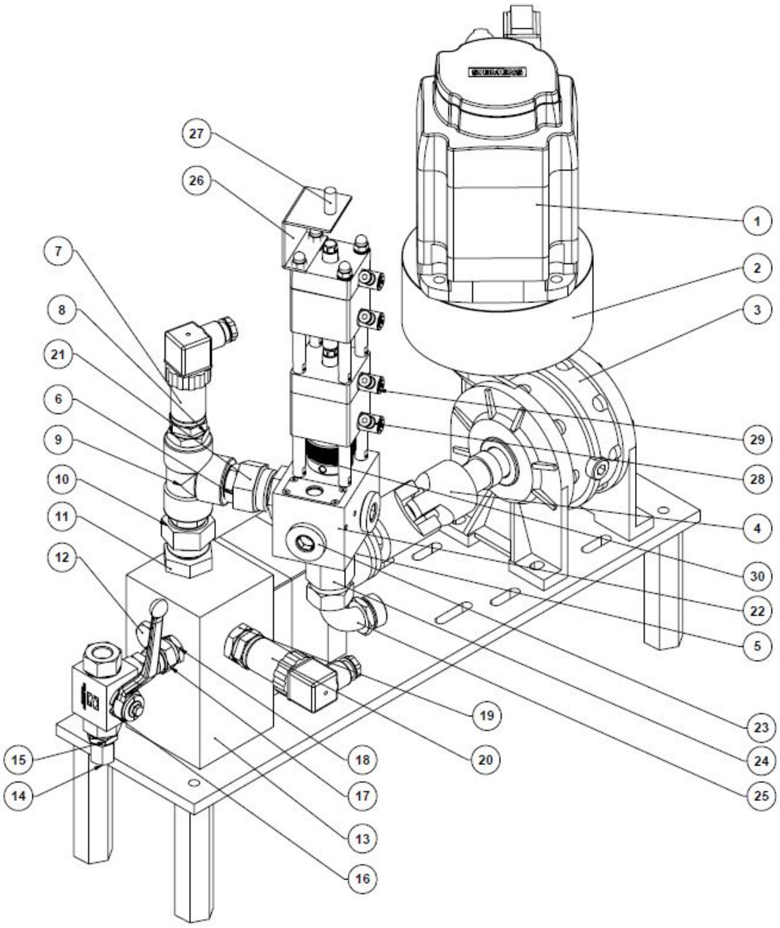
__Results__:

Ambient Temperature : 25.8 °C
RH % : 76.8

Recommended Mix Ratio by weight : 100 : 18
Tolerances : 100 : 16 to 100 : 20

Sample	Weight (g)		Ratio		Remarks	Next action
	Cup A	Cup B	Part A	Part B		
1	119.1	25.5	100.0	21.4	Out of range	Adjust pumps
2	119.2	25.6	100.0	21.5	Out of range	Adjust pumps
3	117.4	25.2	100.0	21.5	Out of range	Adjust pumps
4	120.9	25.8	100.0	21.3	Out of range	Adjust pumps
5	120.3	25.7	100.0	21.4	Out of range	Adjust pumps
6	123.1	23.4	100.0	19.0	Within range but high Part B	Adjust pumps
7	123.0	23.5	100.0	19.1	Within range but high part B	Adjust pumps
8	120.4	21.9	100.0	18.2	Recommended mixing ratio 100:18	-
9	119.3	21.9	100.0	18.4	Recommended mixing ratio 100:18	-
10	121.5	22.2	100.0	18.3	Recommended mixing ratio 100:18	-

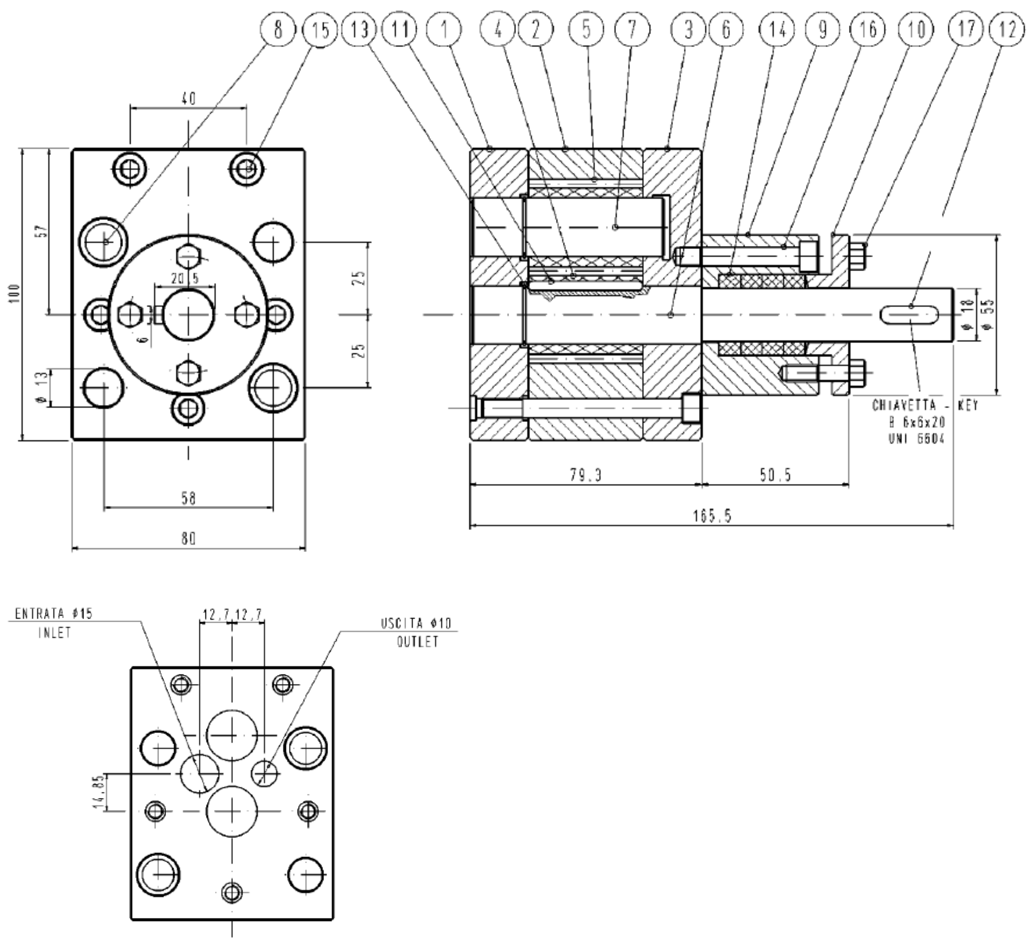
6.8 Gear Pump Dosing System - A Component



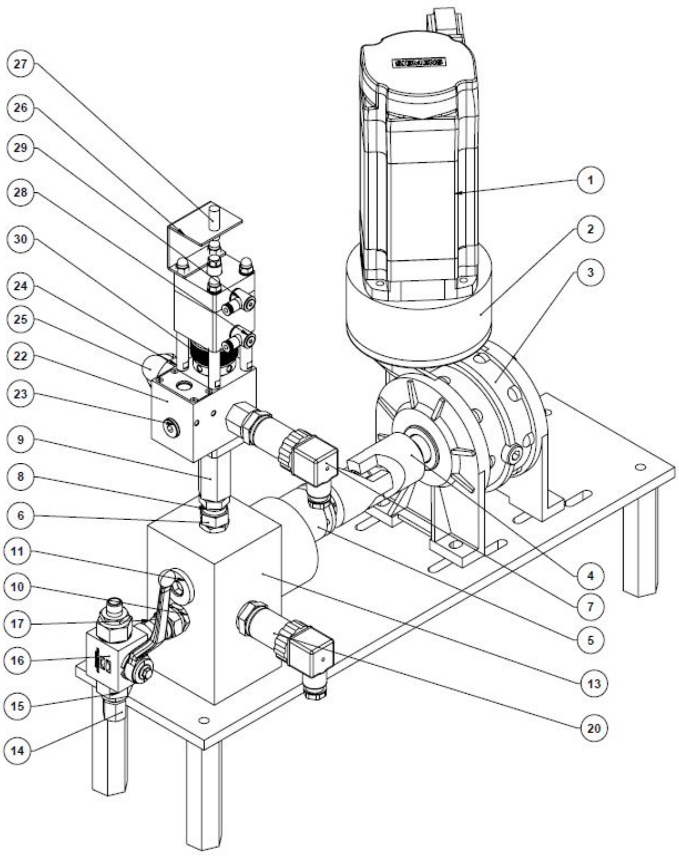
1	Siemens 1.5KW Motor	16	1/2" 3 Way valve
2	Reducing flange	17	1/2" F Union-1/2"M
3	Reducer 40:1 Ratio	18	3/8"M to 1/2"M
4	Coupling	19	3/4"M to 1/2"M Reducer
5	10cc Pump	20	Pressure sensor post gear pump. 1/2"M
6	3/4"M to 3/4"M	21	3/4"F Union to3/4"M
7	Pressure sensor pre gear pump. 1/2"M	22	DMT28 Valve
8	1/2"F to 3/4"M reducer	23	3/4" Cap
9	3/4"F Tee	24	3/4"F to 3/4"M coupling
10	3/4" M to 3/4" M	25	3/4"F to 3/4"M 90Deg Elbow
11	3/4" F Union to Male	26	Valve sensor support
12	1/2" Plug	27	Valve open sensor
13	Manifold 10cc Pump	28	6mm valve air open inlet
14	1/4" Cap	29	6mm valve air close inlet
15	1/2"M to 1/4"M	30	Valve opening regulator

6.9 Gear Pump Dosing System 10CC Pump

1	Internal Plate	10	Presser
2	Gear Plate	11	Shaft Key
3	External Plate	12	Ext Key 6 x 6 x 20mm
4	Driving Gear	13	Circ Clip A20
5	Driven Gear	14	Graphite Braided Gasket 5 x 5 x 500
6	Driving Shaft	15	M6 x 65
7	Driven Shaft	16	M6 x 40
8	Centering Pin	17	M5 x 15
9	Stuffing Box		



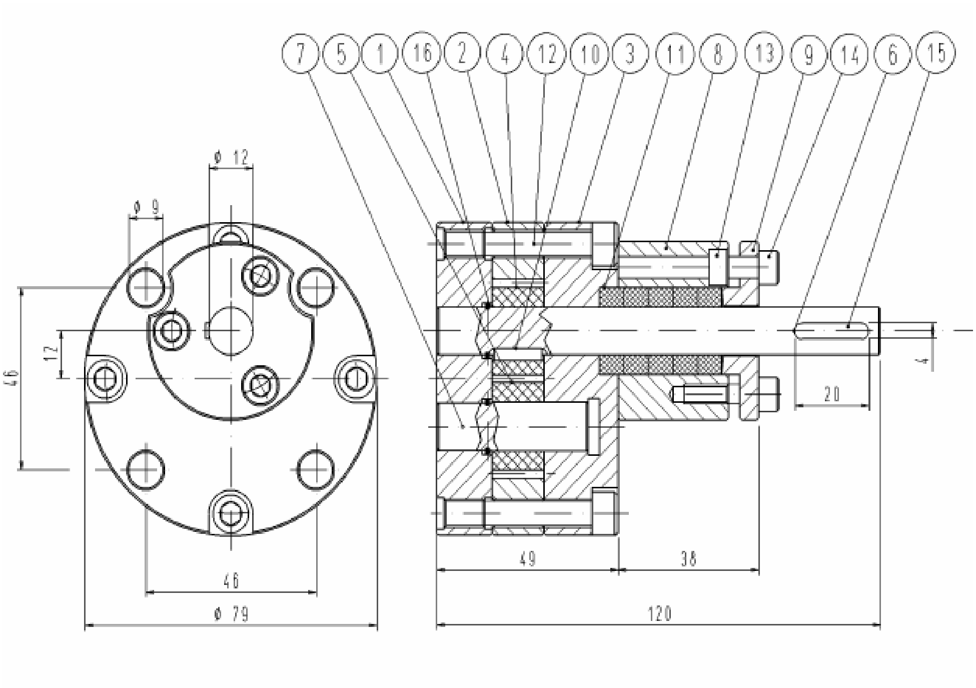
6.10 Gear Pump Dosing System - B Component

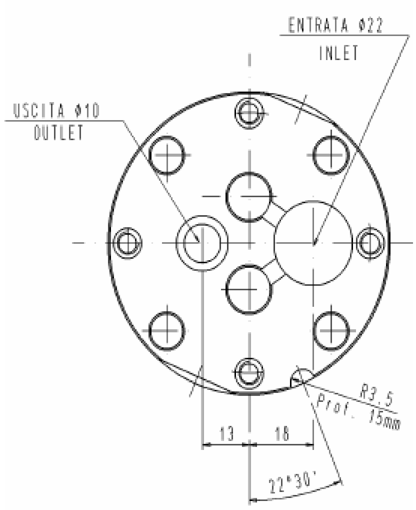


1	Siemens 1KW Motor	16	1/2" 3 Way valve
2	Reducing flange	17	1/2" F Union-1/2"M
3	Reducer 28:1 Ratio	18	
4	Coupling	19	
5	3cc Pump	20	Pressure sensor post gear pump. 1/2"M
6	3/4M to 3/8M coupling	21	
7	Pressure sensor pre gear pump. 1/2"M	22	PU8 3/8
8	3/8"M to 3/8"F Union	23	3/8" Cap
9	Hex 3/8F to 3/8F	24	3/8"M to 3/4"M coupling
10	3/8"M to 1/2" M coupling	25	3/4"F to 3/4"M 90Deg Elbow
11	1/2" Cap	26	Valve sensor support
12		27	Valve open sensor
13	Manifold 3cc Pump	28	6mm valve air open inlet
14	1/4" Cap	29	6mm valve air close inlet
15	1/2"M to 1/4"M	30	Valve opening regulator

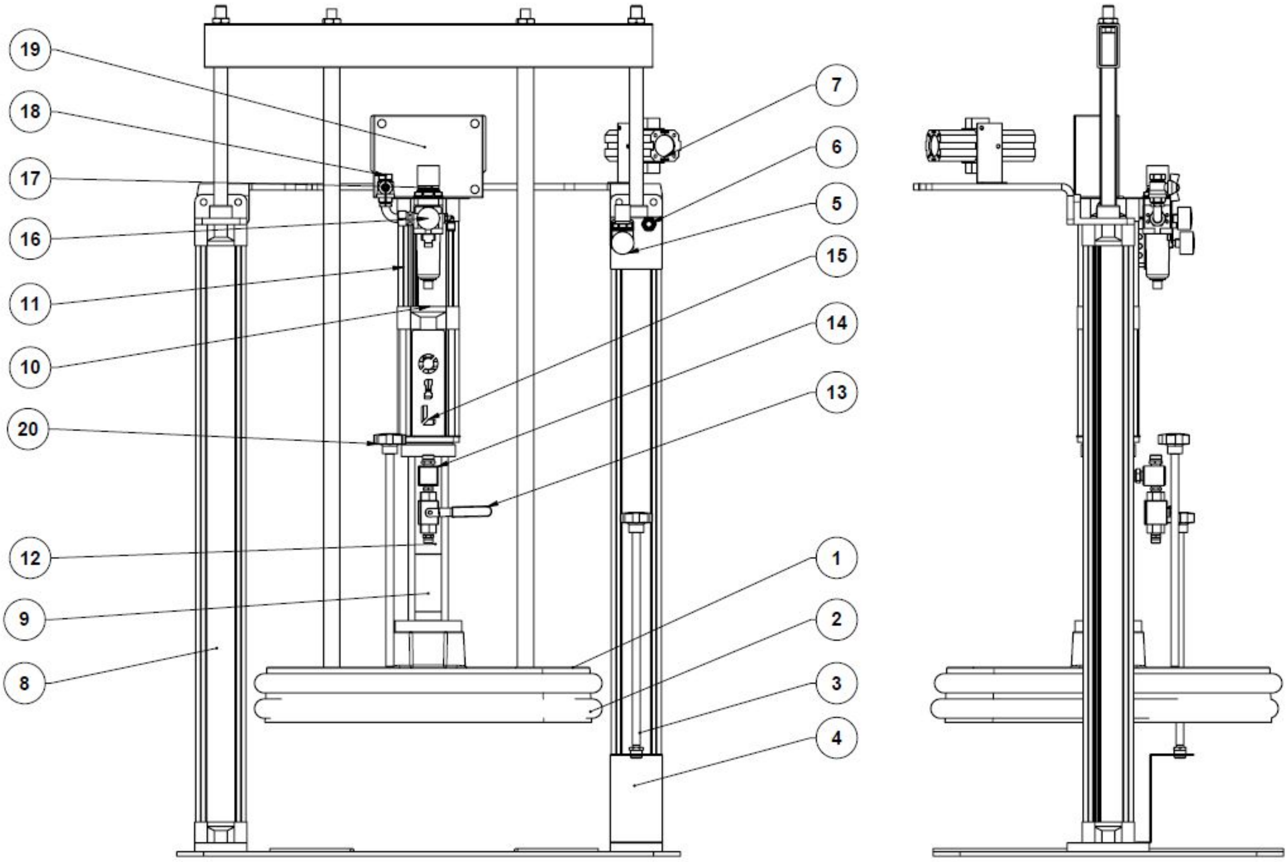
6.11 Gear Pump Dosing System 3CC Pump

1	Internal Plate	9	Presser
2	Gear Plate	10	Key
3	External Plate	11	Gasket 5 x 5
4	Driving Gear	12	Screw M6x40
5	Driven Gear	13	Screw M5x35
6	Driving Shaft	14	Screw M5 x 20
7	Fixed Pin	15	Key type B 4x4x20
8	Stuffing Box	16	Stop ring A12



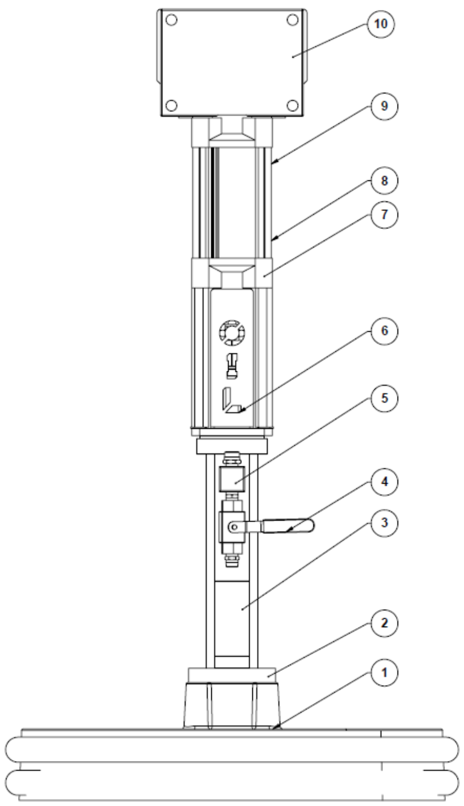


6.12 Transfer Pump - A Component (200Ltr Drum)



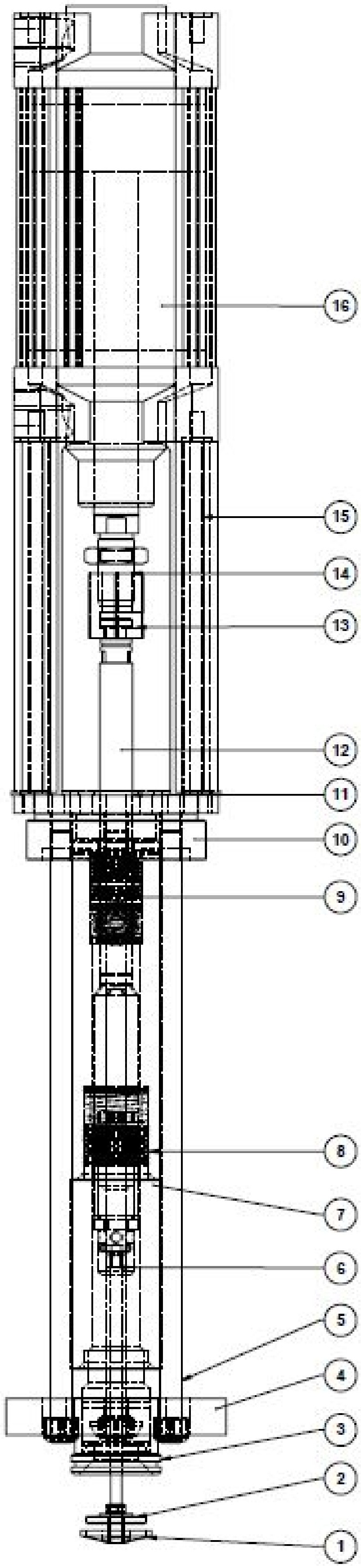
1	Follower Platen
2	EPDM / Silicone O-Ring
3	Air injector Wand
4	Air Injector Wand Support
5	Air pressure regulator and Indicator for Cylinders to raise / lower the platen
6	Level to raise / lower the platen
7	Air pressure booster
8	Pneumatic cylinders to raise / lower the platen
9	Piston pump
10	Air clyinder motor for piston pump
11	Reed switches for cylinder direction switching
12	Glue purge exit
13	3 Way manual purge valve
14	Glue transfer pump exit
15	Seal lubrification point
16	Piston pump pressure indicator (manual version)
17	Piston pump air pressure regulator (manual version)
18	Piston pump air inlet isolation
19	24VDC Electircal box for piston cyclinder sensors
20	Platen purge wand

6.13 Transfer Pump - A Component (200Ltr Drum) – Piston Pump Group



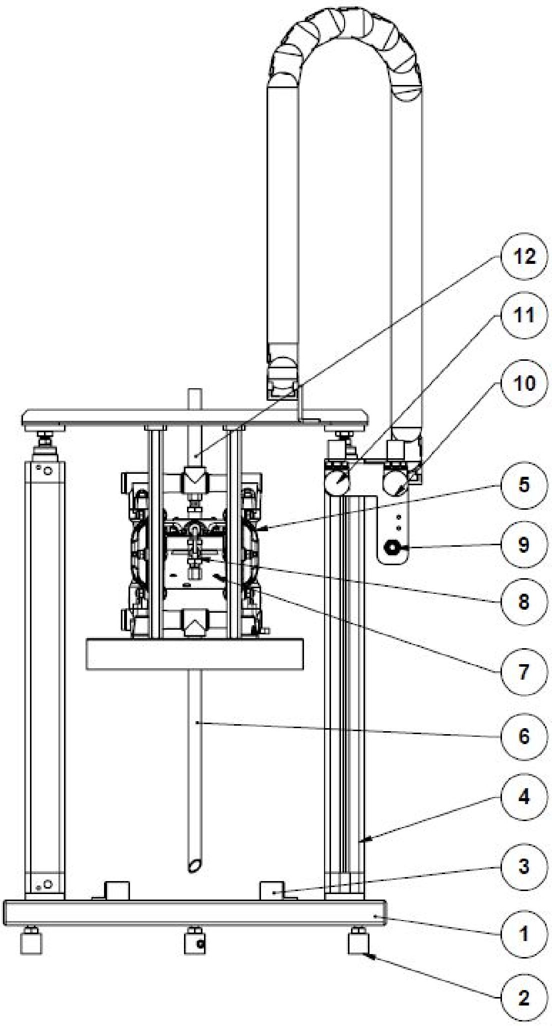
1	Platen.
2	Plate for fixing the piston pump to the platen.
3	Piston Pump.
4	3 Way Valve. Handle in the horizontal position allows the flow to the gear pump, Handle in the vertical position purges the adhesive.
5	Piston pump exit.
6	Oil point at the top of the piston pump. Apply an inert oil such as mesamol to help lubricate the seals and prevent leakage.
7	Pneumatic cylinder to drive the piston up / down.
8	Lower sensor that swithces the pneumatic cylinder to upwards. Note: There are two sensors per stroke direction.
9	Upper sensor that switches the pneumatic cylinder to downwards. Note there are two sensors per stroke direction.
10	Electircal box for connecting the pneuamtic cylinder stroke direction sensors and the main control panel.

6.14 Transfer Pump - A Component (200Ltr Drum) – Piston Pump



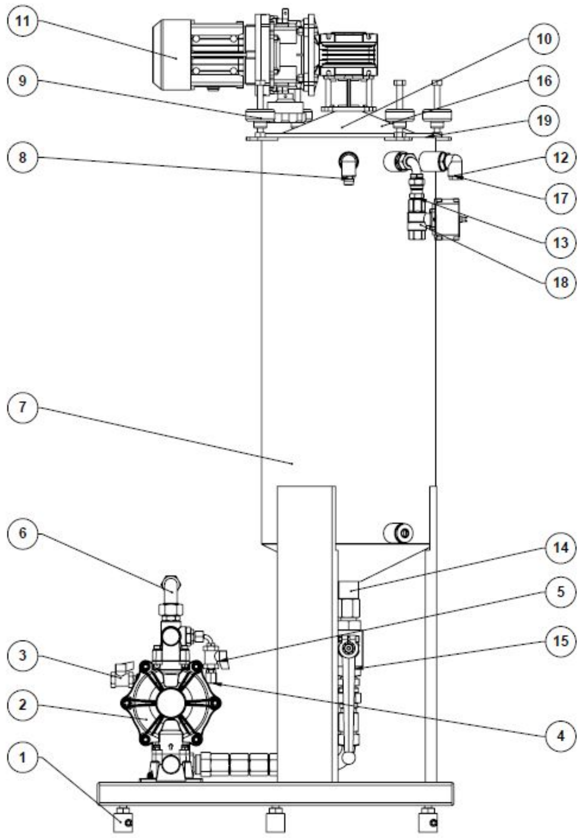
1	Submerging collector plate
2	Floating closing plate
3	Static O-ring seal for piston to Platen joint.l
4	Flange to fix piston to the platen also to hold the piston assembly together.
5	Piston assembly bar, unscrew these to disassemble the inside of the piston.
6	None return valve (sphere and seat)
7	this is the splitting point of the piston main cylinder and contains an o-ring for sealing.
8	This is the block of lower seals for the piston arm.
9	This is the block of upper seals for the piston arm.
10	Upper flange to fix the piston assembly.
11	Oil point for lubrification of the cylinder.
12	Piston arm.
13	Coupling to join the piston arm and the pneumatic cylinder arms
14	Pneumatic cylinder arm.
15	Spacer legs for the coulping zone between the pneumatic cylinder and the piston.
16	Pneematic Cylinder.

6.15 Transfer Pump – B2 Component (20Ltr Drum)



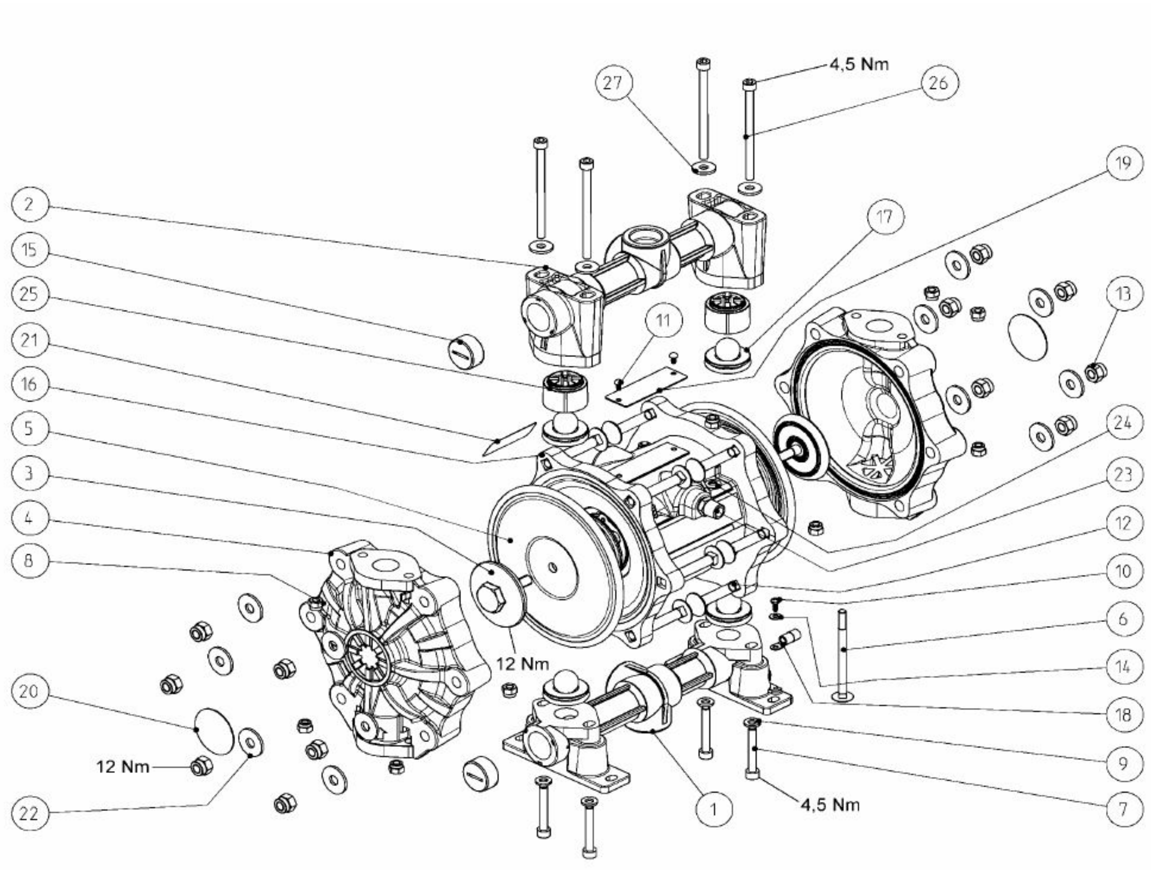
1	Folded Steel Base
2	3 x Load Cells
3	Drum Stop for positioning of drum under the dip pipe
4	Pneumatic Cylinders for raising/lowering the dip pipe into the adhesive drum
5	Air Feed into the Membrane/Diaphragm pump
6	Dip Pipe with self piercing tip
7	membrane / Diaphragm Pump
8	Purge valve
9	Pneumatioc Lever to raise/lower the pump onto the adhesive drum
10	Air regulator for controlling the air pressure to the membrane/diaphragm pump
11	Air regulator to control the air pressure to the cylinders to raise/lower the pump onto the drum
12	Adhesive exit to reservoir tank or to the 3 way valve transfer pump selector valve (configuration dependent)

6.16 Transfer Pump – B1 Component (Reservoir tank)



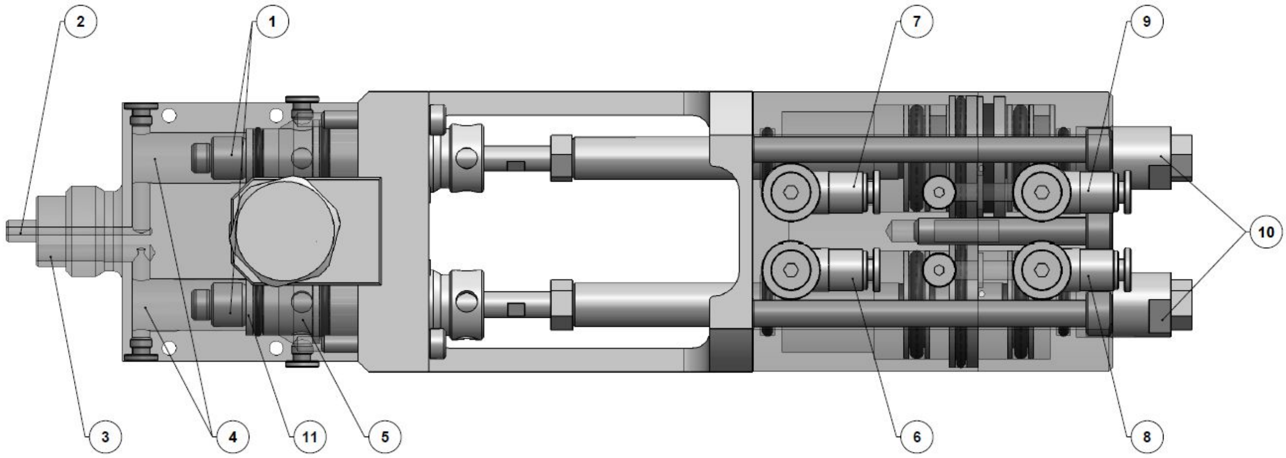
1	Load Cells for measuring the weight of material in the tank (necessary to tare the components also on the balance).
2	DM80 Membrane/Diaphragm transfer pump to the soing gear pump.
3	Transfer pump air inlet and air supply isolation valve.
4	Transfer pump adhesive purge cap. Necessary to keep in place when not in use to prevent adhesive curing. Reccommend lubrictaing with inorganic grease.
5	Transfer pump manual adhesive purge valve.
6	Transfer pump exit to dosing pump via hose.
7	B1 B Component Reservoir Tank.
8	Dray Air exit pressure relief valve.
9	Reservoir tank fixings.
10	Reservoir tank removable lid. Note: Should remain frimly in place when in use to prevent contamination with moisture from air.
11	Motor for adhesive agitation. Manual speed adjustment via reducer handle (optional extra).
12	Dry air inlet to create dry air blanket over the adhesive surface. Sensor for measuing air present from dry air unit and over pressure alarm.
13	Reservoir tank adhesive filling inlet from B2 drum transfer pump.
14	Reservoir tank outlet.
15	Manual 2 way isoltation valve for adhesive exit from reservoir tank.
16	Laser level sensor. Secondary measure for glue level to prevent over filling in the case of load cell failure (Optional extra).
17	Air pressure sensor to detect low / high dry air pressure from the dry air unit, or dry air unit failure (optional extra).
18	Two valve with penumatic actuator for controlling the feeding of the adhesive into the reservoir tank from the B2 drum transfer pump.
19	Disposable Bag System for easy cleaning of the tank(optional extra).

6.17 Transfer Pump - B Component Membrane Pump component List



FINDIEMME ADHESIVE TECHNOLOGIES

6.18 Applicator Head – EziMix



Important

The Ezi-Mix movement is down to open and up to close.

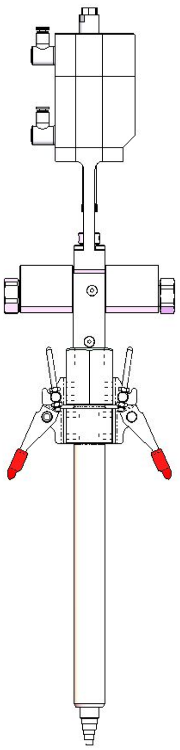
For the “Ezi-Mix+” version there is an additional piston to allow the A & B air cylinders to be linked together during extrusion and free to move independently when enabled individually. This ensures both A & B open simultaneously for every extrusion regardless of dissimilar pressures in the line.

An additional function of the “Ezi-Mix+” is that there are reed sensors to read the forward/back position of the pneumatic cylinder.

1	Plastic Seat that seals the glue feed when closed
2	B Component Exit (smaller to balance pressure)
3	A Component Exit (set back to redeuce cross contamination)
4	Glue channel
5	Glue seal cartridge
6	A Component Air supply close
7	B Component Air supply close
8	A Component Ar supply Open
9	B Component Air Supply Open
10	Registration of the gun open movement
11	Seat for closing the gun

6.19 Applicator Head: Ezi-Mix Quick Lock (optional)

Image 1



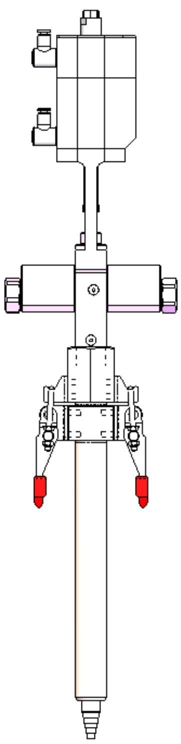


Image 2

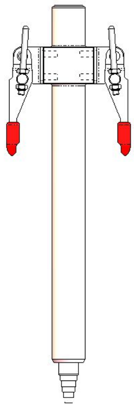


Image 3

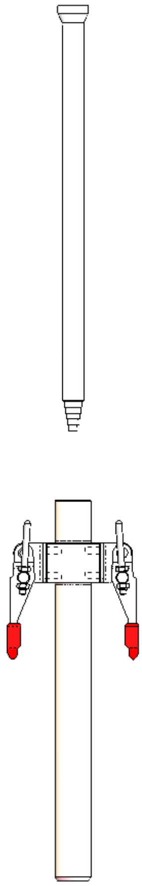


Image 4

To change the mixer nozzle when using the “Quick Lock” system.

1- Lift the levers shown in red (image 1)
2- Remove the mixer holder (image 2)
3- Remove the old mixer and replace with a brand new mixer element (image 3).
4- Replace the mixer holder and engage the levers to clamp the mixer in place.

If the adhesive leaks during extrusion tighten the nuts on the clamp levers enough to allow them to be engaged easily whilst still creating a perfect seal.

This system is designed for mixer type MS10-24.

7. Software:

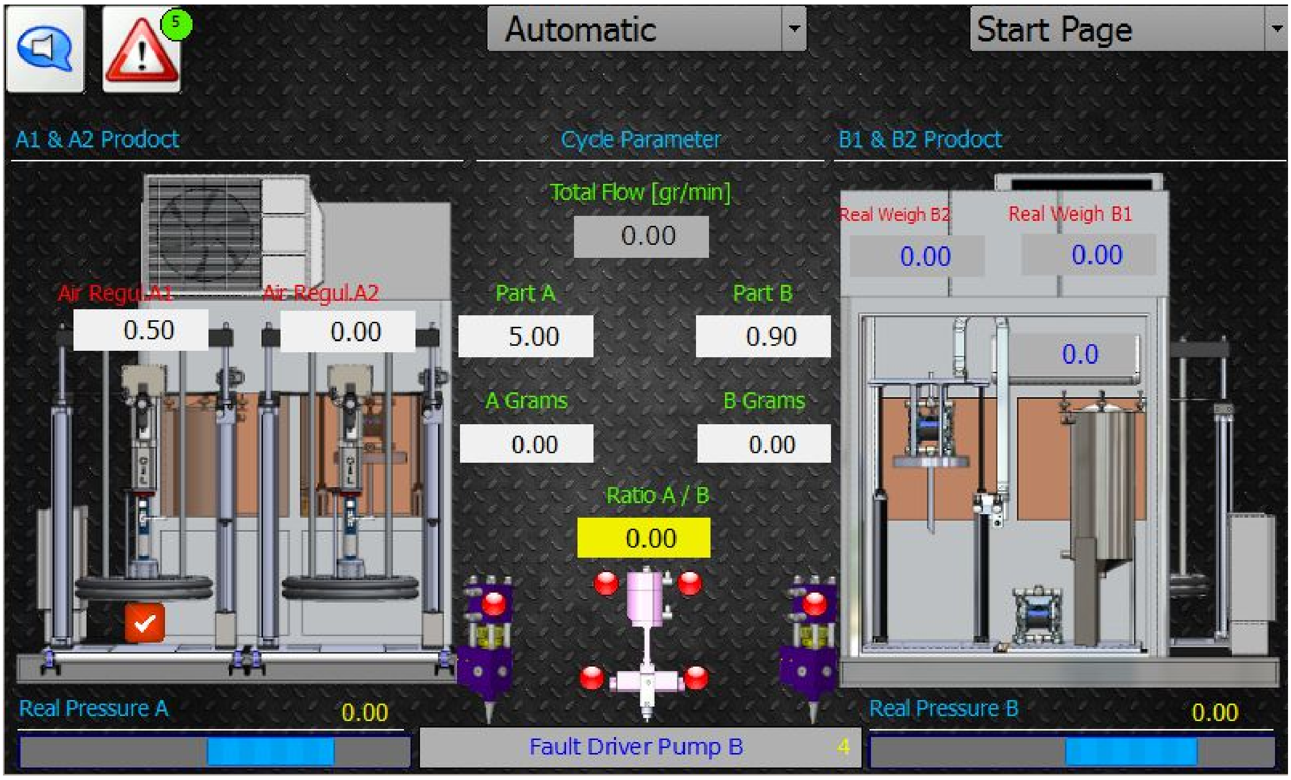
7.1 AUTOMATIC SCREEN - LAYOUT

This is the default screen when the PLC loads and is the main working page for the operator. It can additionally be selected from the centre drop-down menu when navigating through all the operator screens.

Within the “Automatic” screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

“Layout” is the main “Automatic” screen which shows an overview of the complete workings of the equipment and the process data.

The principle purpose of the “Automatic” ambient is for working when integrated as part of a glue cell with an external start/stop signal.

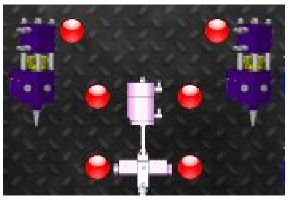


Descriptions of display view starting from left, top to bottom:

- **Drums Product A1 and A2:** Title indicating the values below are for managing component A.
- **Air Pressure A1**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 1.
- **Air Pressure A2**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 2.
- **Pressure A / B**: Pressure indication for component A and B in Bar prior to the PU8 two way valve.
- **Automatic**: Shows that the “Automatic” ambient has been selected form the central drop down menu. This is the normal ambient for working together with the signal from the robot.
- **Start Page**: Shows that “start page” ambient has been selected from the right drop down menu, this displays an overview of the full machine.

- **Data and Action:** Title indicating the values below are the process data for the extrusion.
- **Total Flow (gr/min)**: This value is the required total flow of A & B combined and expressed in grams per minute.
- **Ratio A:** This value indicates the ratio of A component required by Weight.
- **Ratio B:** This value indicates the ratio of B component required by Weight.
- **Total A (gr):** This values displays the total quantity of A component dosed during the last extrusion in grams.
- **Total B (gr):** This values displays the total quantity of B component dosed during the last extrusion in grams.
- **Drums Product B1 and B2:** Title indicating the values below are for managing component B.
- **Weight B2:** Weight of the adhesive in the 20Ltr B component Transfer pump.
- **Weight B1:** Weight of the adhesive in the B component 50Ltr Tank.
- **Pressure PU8 – B:** Pressure indication for component B expressed in Bar prior to the PU8 two way valve.

- **Pressure Easy Mix B:** Pressure indication for component B expressed in Bar at the gear pump exit.



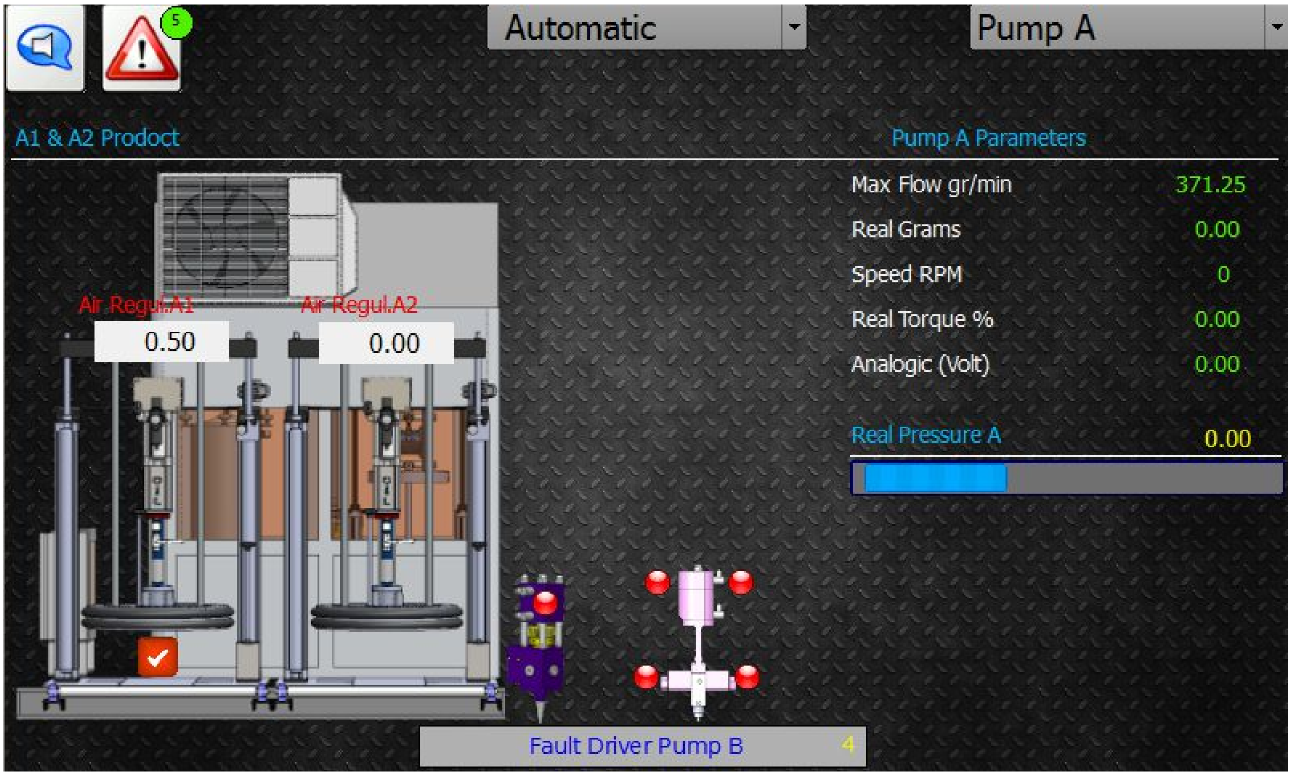
The Circular Lights (seen in the above image in red) indicate the position of the valves.
The PU8 two way valves prior to the gear pump will display green when open and red when closed.
The Extrusion gun (Ezi-mix) has two pneumatic pistons to allow A and B components to be open independently. When the display turns green this is the position, the upper light being green signifies it is closed and the lower light signifying it is open.

The green light at the bottom right of the PF200 (A Component 200Ltr transfer pumps) changes to red when the low level sensor has been enabled indicating that the drum needs to be changed (Image above)

7.2 **AUTOMATIC SCREEN – PRODUCT A**

Within the "Automatic" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Product A" shows an overview of the A Component process data:



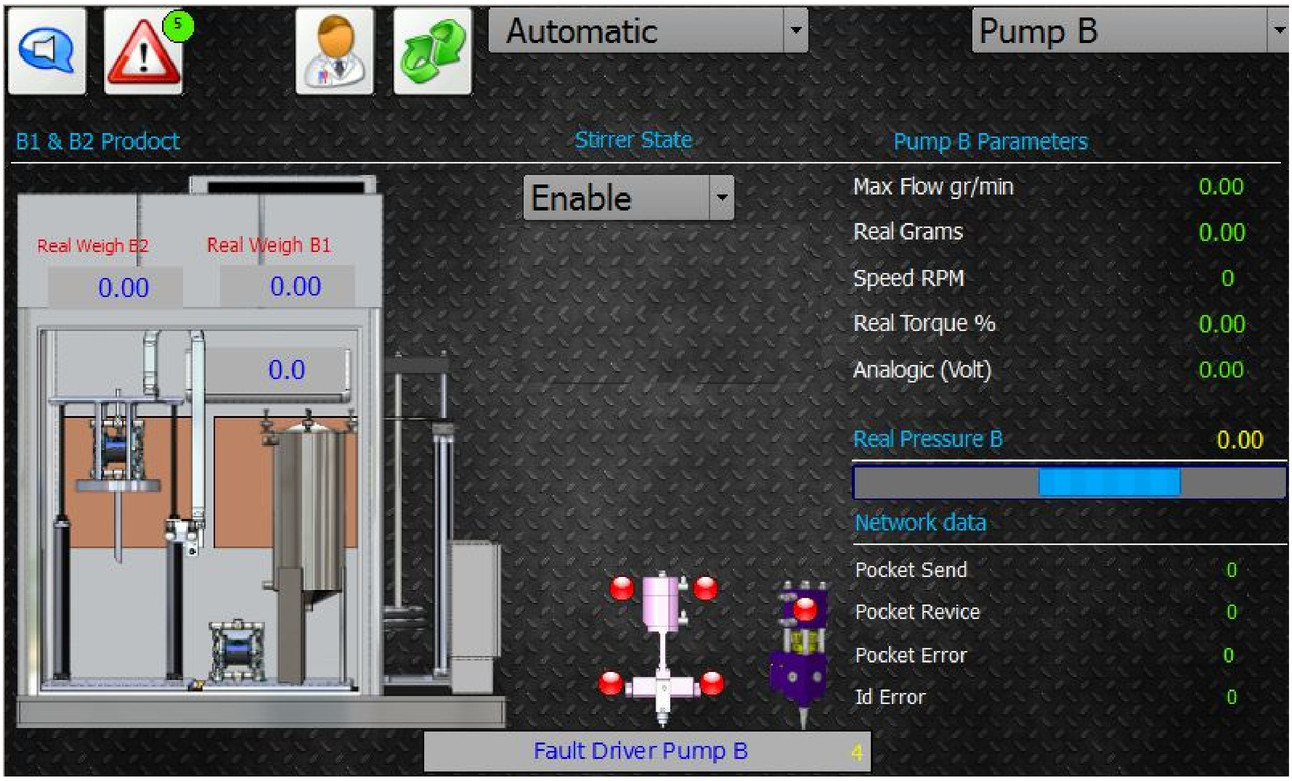
- **Drums Product A1 and A2:** Title indicating the values below are for managing component A.
- **Air Pressure A1**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 1.
- **Minimum level A1:** The green light at the bottom right of the PF200 (A1omponent 200Ltr transfer pumps) changes to red when the low level sensor has been enabled indicating that the drum needs to be changed (Image above).
- **Air Pressure A2**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 2.
- **Minimum level A2:** The green light at the bottom right of the PF200 (A2 Component 200Ltr transfer pumps) changes to red when the low level sensor has been enabled indicating that the drum needs to be changed (Image above)
- **Parameters:** Title indicating that the values below are for the A component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the A Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component A gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.

- **In Pressure PV8 – A:** This displays the pressure in Bars at the PU8 two way valve from the PF200 transfer pump prior to the gear pump

7.3 AUTOMATIC SCREEN – PRODUCT B

Within the "Automatic" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

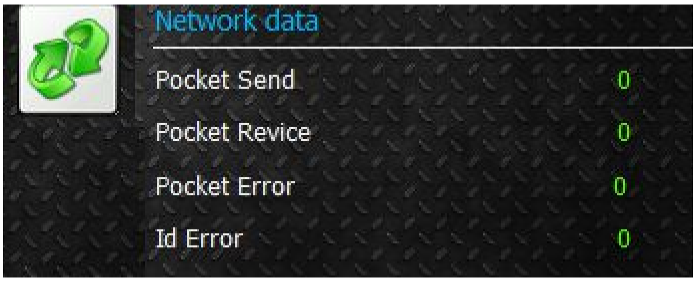
"Product B" shows an overview of the B Component process data:



- **Drums Product B1 and B2:** Title indicating the values below are for managing component B.
- **Load Cells B2:** The value below this title indicates the weight in Kilograms remaining inside the B Component 20 Ltr drum.
- **Load Cells B1:** The value below this title indicates the weight in Kilograms remaining inside the B Component reservoir tank.
- **Parameters:** Title indicating that the values below are for the B component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the B Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component B gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.
- **Pressure PV8 – B:** This displays the pressure in Bars at the PU8 two way valve from the B1 reservoir tank membrane transfer pump prior to the gear pump
- **Real Weight Cells B1:** Indicates the real weight on the Tank B1 (Kg)
- **Real Weight Cells B2:** Indicates the real weight on the Drum B2 (Kg)

In all screens the “Attention” sign in the top right of the screen indicates there is an alarm in the working of the equipment. When the alarm is cleared it will either automatically vanish or by pressing the alarm to accept it, the machine will clear the alarm.

The “red tick box” to the top right of the screen indicates…….



If they press refresh button, reset the Modbus network from plc to load Cells.

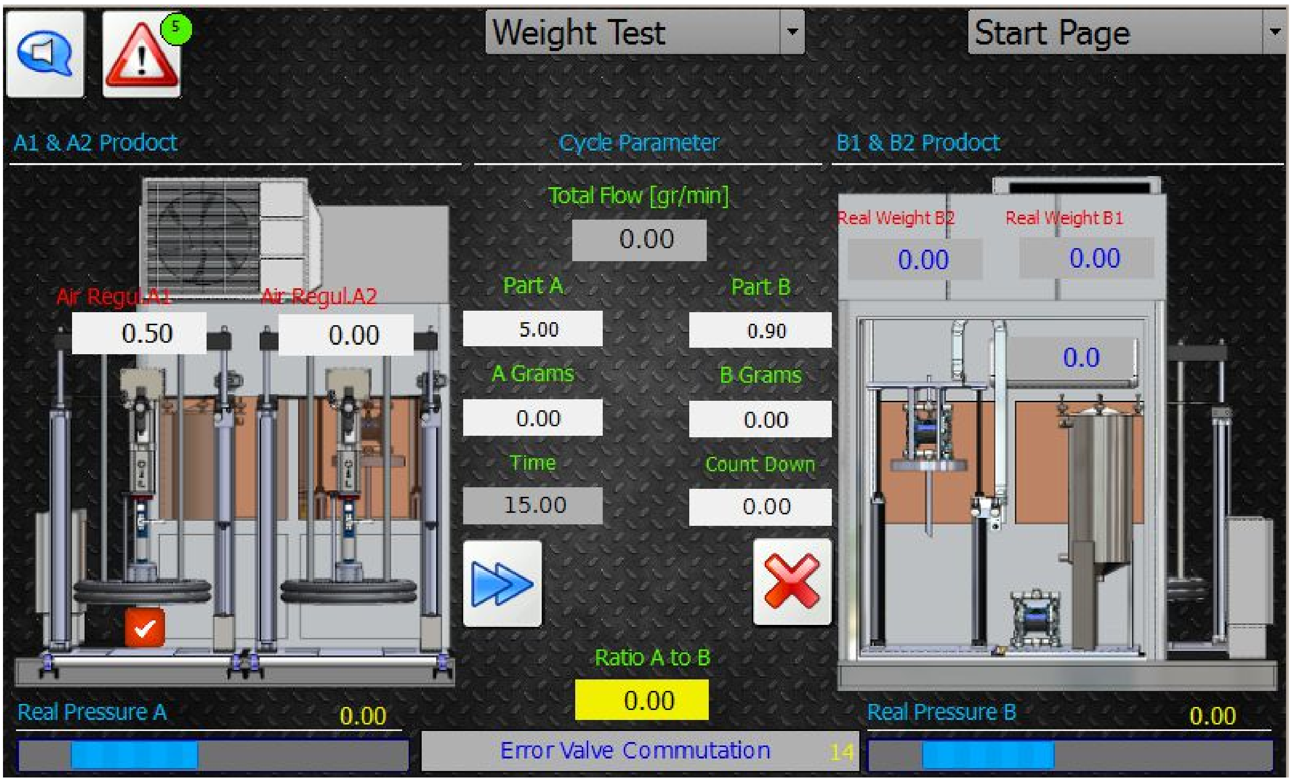
- Pocket Send : Number of the request to Plc from the slave Node (Load cells and / or massic Flow).
- Pocket Recive : Number of pocket recive from slave Node (Load cells and / or massic Flow)to Plc.
- Pocket Error : Number of pocket Error).
- Id error: Last Slave Id generate the error on network.

7.5 WEIGHT TEST - LAYOUT

The weight test screens can be selected by first selecting “Weight Test” from the centre drop-down menu and then navigating through the various options in the right drop down menu.

“start page” is the main “Weight Test” screen which shows an overview of the complete workings of the equipment and the process data:

The principle purpose of the “Weight test” ambient is to allow manual extrusion for the set-up, calibration and repeatability testing of the dosing prior to production.



Descriptions of display view starting from left, top to bottom:

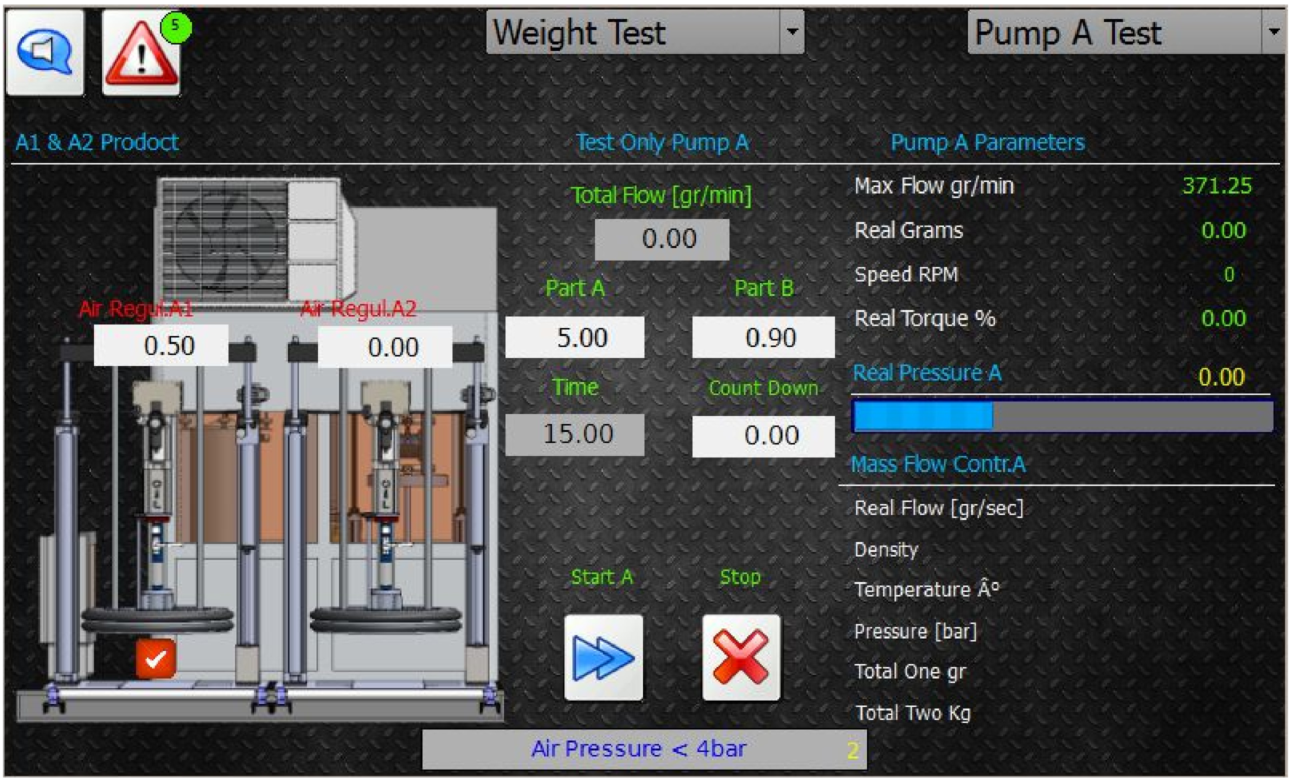
- **Drums Product A1 and A2:** Title indicating the values below are for managing component A.
- **Air Pressure A1**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 1.
- **Air Pressure A2**: Value in Bar of the air pressure supply for driving the piston pump on the drum transfer pump A component, drum 2.
- **Pressure PU8 – A**: Pressure indication for component A in Bar prior to the PU8 two way valve.
- **Pressure Easy Mix A**: Pressure indication for component A in Bar at the gear pump exit.
- **Weight Test**: Shows that the “Weight test” ambient has been selected form the central drop down menu. This is the normal ambient for working in a manual mode.
- **Layout**: Shows that “Layout” ambient has been selected from the right drop down menu, this displays an overview of the full machine.
- **Data and Action:** Title indicating the values below are the process data for the extrusion.
- **Total Flow (gr/min**): This value is the required total flow of A & B combined and expressed in grams per minute.
- **Ratio A:** This value indicates the ratio of A component required by Weight.

- **Ratio B:** This value indicates the ratio of B component required by Weight.
- **Time Test:** Expressed in milliseconds this displays the time that both A & B pumps will extrude adhesive when the "Start All Pump" button is pressed.
- **Countdown:** Shows a real time count down of the time set in "Time Test" when the "start All Pump" button is pressed.
- **Start All Pump:** Push this button to start the extrusion. It will start both gear pumps for the period set in "Time Test".
- **Stop All Pump:** If you wish to stop the gear pumps extruding product prior to the countdown finishing push this button.
- **Drums Product B1 and B2:** Title indicating the values below are for managing component B.
- **Weight B2:** Weight of the adhesive in the 20Ltr B component Transfer pump.
- **Weight B1:** Weight of the adhesive in the B component 50Ltr Tank.
- **Pressure PU8 – B:** Pressure indication for component B expressed in Bar prior to the PU8 two way valve.

7.6 WEIGHT TEST – TEST WEIGHT A

Within the "Weight Test" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Test Weight A" shows an overview of the A Component process data:



- **Drums Product A1 and A2:** Title indicating the values below are for managing component A.
- **Pressure PU8 – A**: Pressure indication for component A in Bar prior to the PU8 two way valve.
- **Weight Test**: Shows that the "Weight test" ambient has been selected form the central drop down menu. This is the normal ambient for working in a manual mode.

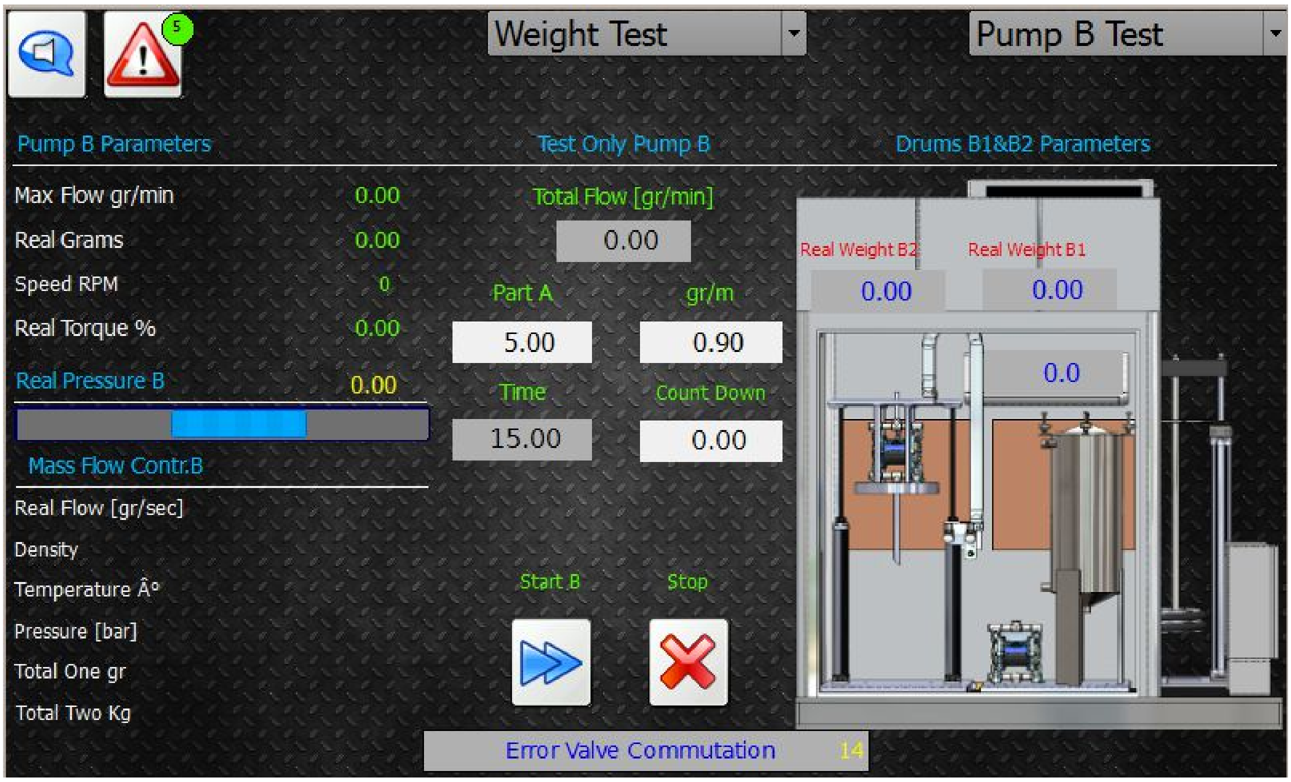
- **Pump Weight A**: Shows that “Test Weight A” ambient has been selected from the right drop down menu, this displays an overview of the full machine.
- **Test only A:** Title indicating the values below are the process data for the extrusion.
- **Total Flow (gr/min**): This value is the required total flow of A & B combined and expressed in grams per minute.
- **Ratio A:** This value indicates the ratio of A component required by Weight.
- **Ratio B:** This value indicates the ratio of B component required by Weight.
- **Time Test:** Expressed in milliseconds this displays the time that both A & B pumps will extrude adhesive when the “Start All Pump” button is pressed.
- **Countdown:** Shows a real time count down of the time set in “Time Test” when the “start All Pump” button is pressed.
- **Start All Pump:** Push this button to start the extrusion. It will start both gear pumps for the period set in “Time Test”.
- **Stop All Pump:** If you wish to stop the gear pumps extruding product prior to the countdown finishing push this button.

- **Parameters:** Title indicating that the values below are for the A component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the A Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component A gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.
- **Massic Flow Pump A:** Title indicating that the values below are from the mass flow meter related to this gear pump.
- **Real Flow (gr/min):** Real time flow from this gear pump expressed in grams per minute.
- **Volume Flow:** Volumetric flow from this gear pump.
- **Density:** The material density traveling through this mass flow meter.
- **Temperatur**e: The material temperature traveling though this mass flow meter.
- **Total 1 (gr):**
- **Total 2 (kg):**

7.7 WEIGHT TEST – TEST WEIGHT B

Within the "Weight Test" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Test Weight B" shows an overview of the B Component process data:



- **Parameters:** Title indicating that the values below are for the B component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the B Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component B gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.
- **Massic Flow Pump B:** Title indicating that the values below are from the mass flow meter related to this gear pump.
- **Real Flow (gr/min):** Real time flow from this gear pump expressed in grams per minute.
- **Volume Flow:** Volumetric flow from this gear pump.
- **Density:** The material density traveling through this mass flow meter.
- **Temperatur**e: The material temperature traveling though this mass flow meter.
- **Total 1 (gr):**
- **Total 2 (kg):**

- **Test Only B :** Title indicating the values below are the process data for the extrusion.
- **Total Flow (gr/min**): This value is the required total flow of A & B combined and expressed in grams per minute.
- **Ratio A:** This value indicates the ratio of A component required by Weight.
- **Ratio B:** This value indicates the ratio of B component required by Weight.
- **Time Test:** Expressed in milliseconds this displays the time that both A & B pumps will extrude adhesive when the "Start All Pump" button is pressed.
- **Countdown:** Shows a real time count down of the time set in "Time Test" when the "start All Pump" button is pressed.
- **Start All Pump:** Push this button to start the extrusion. It will start both gear pumps for the period set in "Time Test".
- **Stop All Pump:** If you wish to stop the gear pumps extruding product prior to the countdown finishing push this button.

- **Drums Product B1 and B2:** Title indicating the values below are for managing component B.
- **Pressure PU8 – B:** Pressure indication for component B expressed in Bar prior to the PU8 two way valve.

7.8 WEIGHT TEST – CAPABILITY TEST

Within the "Weight Test" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Capability Test" gives an overview of both A & B gear pumps whilst carrying out repeated extrusions the timings of which can be set and the values from the flow meters compared to those taking physically by a technician.

- **Parameters Pump A:** Title indicating that the values below are for the A component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the A Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component A gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.
- **Massic Flow Pump A:** Title indicating that the values below are from the mass flow meter related to this gear pump.
- **Real Flow (gr/min):** Real time flow from this gear pump expressed in grams per minute.
- **Volume Flow:** Volumetric flow from this gear pump.
- **Density:** The material density traveling through this mass flow meter.
- **Temperatur**e: The material temperature traveling though this mass flow meter.
- **Total 1 (gr):**
- **Total 2 (kg):**

- **Total Flow (gr/min**): This value is the required total flow of A & B combined and expressed in grams per minute.
- **Ratio A:** This value indicates the ratio of A component required by Weight.
- **Ratio B:** This value indicates the ratio of B component required by Weight.
- **Time Test:** Expressed in milliseconds this displays the time that both A & B pumps will extrude adhesive when the "Start All Pump" button is pressed.
- **Countdown:** Shows a real time count down of the time set in "Time Test" when the "start cycle" button is pressed.
- **Time Test/Countdown:** Press to enable testing cycle. When "On" is displayed test is in progress, when "Off" is displayed test is cycle is disabled.
- **Time Wait Test:** Expressed in milliseconds this displays the delay time between each extrusion of the cycle.
- **Countdown:** Shows a real time count down of the time set in "Time Wait Test" when the "start cycle" button is pressed.
- **Number of Test want:** Insert here the number of cycles desired. One cycle consists of the "Time test + Time Test Wait".
- **Test OK:** Displays which cycle is running.

- **Parameters Pump B:** Title indicating that the values below are for the B component Gear Pump
- **Max Flow (gr/min):** This displays the maximum possible grams per minute pump rate of the B Components gear pump based on the pump set up.
- **Real Grams (gr/min):** This displays the real pump quantity of Component B gear pump in grams per minute
- **Actual Speed (rpm):** This displays the data feedback from the motor driver and shows the real time rotation per minute of the motor.
- **Actual Torque %:** This displays the data feedback from the motor driver and shows the real time percentage torque of the motor driver.

- **Massic Flow Pump B:** Title indicating that the values below are from the mass flow meter related to this gear pump.
- **Real Flow (gr/min):** Real time flow from this gear pump expressed in grams per minute.
- **Volume Flow:** Volumetric flow from this gear pump.
- **Density:** The material density traveling through this mass flow meter.
- **Temperature**: The material temperature traveling though this mass flow meter.
- **Total 1 (gr):**
- **Total 2 (kg):**

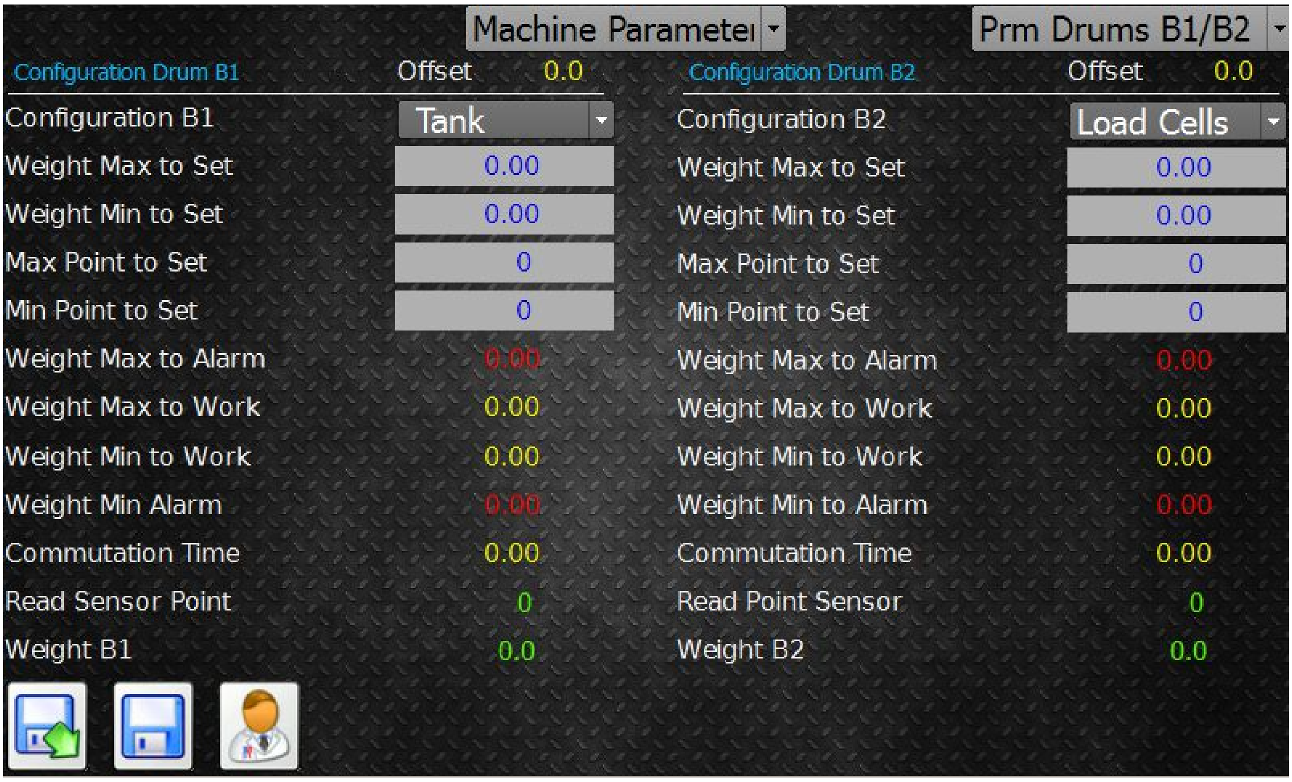
7.9 HISTORY ALARM

From the centre drop down menu you can select "History Alarm". This displays a historical list of alarms that have taken place.

7.11 SETTING MACHINE – LOAD CELLS B1 AND B2

Within the "Setting Machine" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Load Cells B1 and B2" shows an overview of the system controlling the weight of Component B in both the 20Ltr drum unloading system and the reservoir tank system:



- **Configuration Drums B1:** Title indicating that all the values below are related to the B1 reservoir tank system.
- **Weight Max to Alarm:** This weight value (Kg) will create a high weight level alarm when reached, stopping further production cycles. Machine will be in alarm.
- **Weight Max to Work:** This weight value (Kg) is the level that you wish to fill the tank to.
- **Weight Min to work:** The weight value (Kg) is the level in which the tank will start to fill from B2.
- **Weight Min to Alarm:** This weight value (Kg) will create a low weight level alarm when reached, stopping production the cycles. Machine will be in alarm.
- **Real Weight (Kg):** Displays the real weight value in Kg for B1.

The B1 tank automatically maintains a level between the values input in "Weight Max to Work" and "Weight Min to work".
When "Weight Min to Work" is reached the system will call glue from the B2 transfer pump until the "Weight Max to work" is reached. The adhesive will be dosed and when the "Weight Min to Work" is reached it will begin refilling again to the "Weight Max to work". The equipment automatically cycles between these two values.

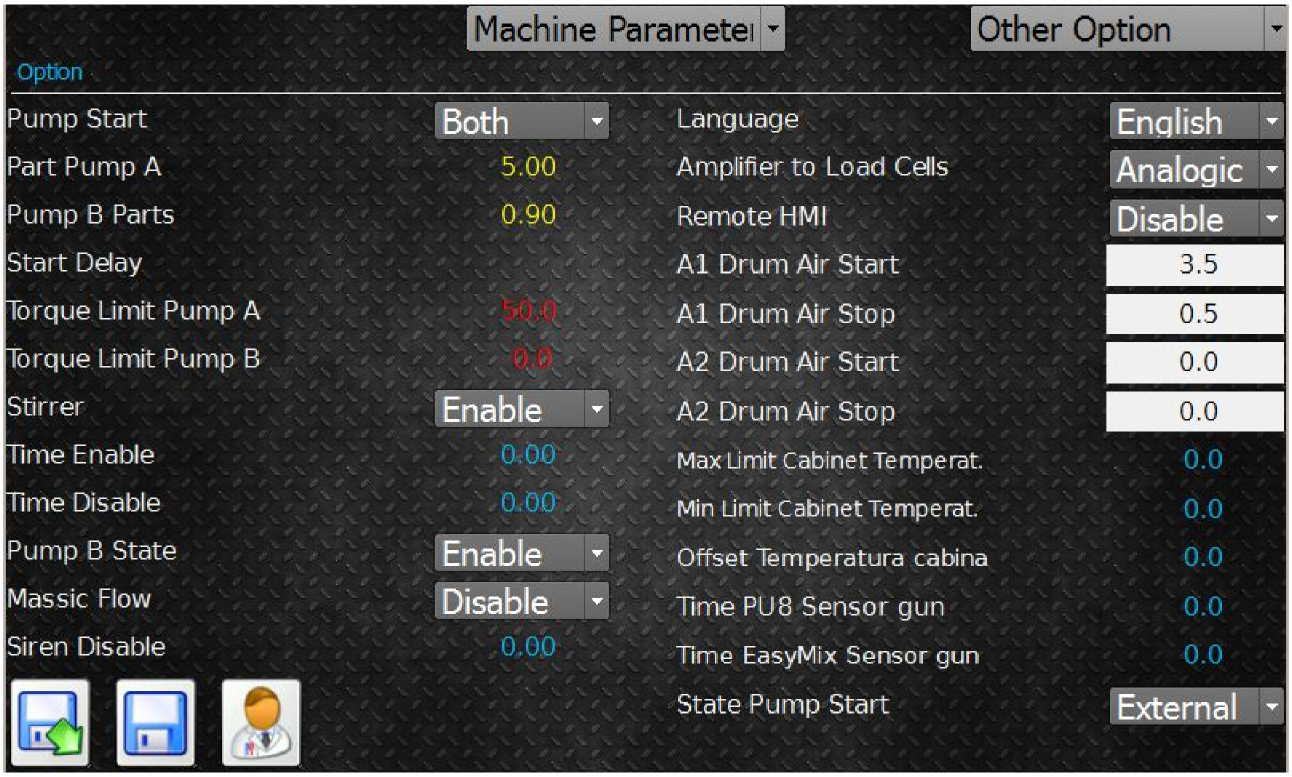
- **Configuration drums B2:** Title indicating that all the values below are related to the B2 drum transfer system.

- **Weight Max to Alarm:** This weight value (Kg) will create a high weight level alarm when reached, stopping further production cycles. Machine will be in alarm.
- **Weight Max to Work:** This weight value (Kg)…..
- **Weight Min to work:** The weight value (Kg) is the low level alarm indicating that the drum is empty and needs to be changed. The equipment will display an alarm but it will allow production cycles to continue.
- **Weight Min to Alarm:** This weight value (Kg)…….
- **Real Weight (Kg):** Displays the real weight value in Kg for B1.
-

7.12 SETTING MACHINE – OTHER OPTION

Within the "Setting Machine" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Load Cells B1": Additional control parameters include:



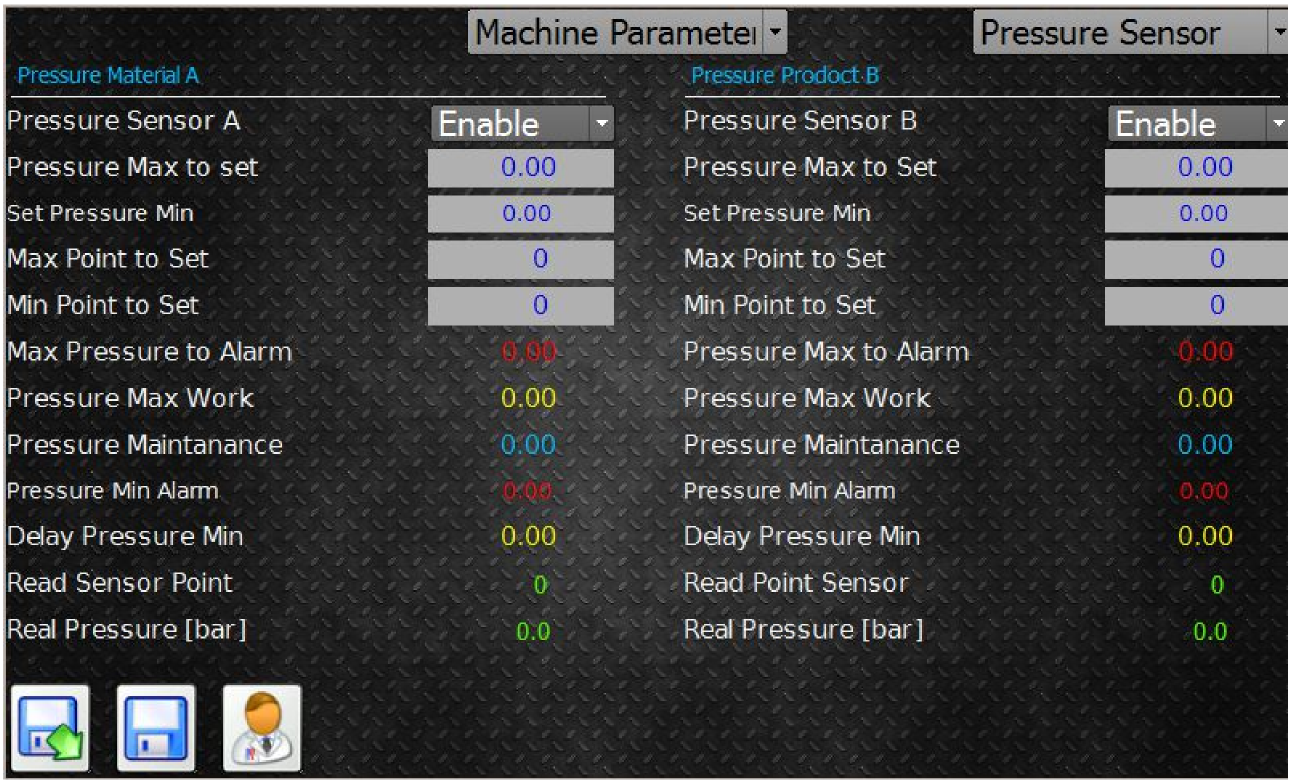
- **Air Pressure A1 start:** Expressed in Bar this value controls the air pressure applied to the A1 piston pump on the PF200 during extrusion.
- **Air Pressure A2 start:** Expressed in Bar this value controls the air pressure applied to the A2 piston pump on the PF200 during extrusion.
- **Air Pressure A1 stop:** Expressed in Bar this value controls the air pressure applied to the A1 piston pump on the PF200 when the machine is idle.
- **Air Pressure A2 stop:** Expressed in Bar this value controls the air pressure applied to the A2 piston pump on the PF200 when the machine is idle.
- **Time PU8 Sensor Gun :** Expressed in seconds, this is the tolerance value permitted for the time it takes for the A and B component cylinder of the applicator head (PU8 Gun) to open.

- **Time Control Easy Mix :** Expressed in seconds, this is the tolerance value permitted for the time it takes for the A and B component cylinder of the applicator head (Ezi-Mix) to open and close.

7.13 SETTING MACHINE – SENSOR PRESSURE PUMP

Within the "Setting Machine" screen there are a number of screens that can be selected from the drop-down menu situated on the top right of the screen.

"Sensor Pressure Pump" encompasses all the data to calibrate the pressure sensors between the gear pump and the applicator head:



- **Sensor Pressure Pump A:** Title indicating that all below values are related to the pressure sensor situated on the exit of the A Component gear pump.
- **Pressure Max to Set:** Upper pressure value expressed in Bar to calibrate the pressure sensor.
- **Set Pressure Min:** Lower pressure value expressed in Bar to calibrate the pressure sensor.
- **Point Maximum Set :** Upper value expressed in points that represents the analogue signal received from the pressure sensor relating to the pressure input in the parameter "Pressure High Set Point".
- **Point Minimum Set :** Lower value expressed in points that represents the analogue signal received from the pressure sensor relating to the pressure input in the parameter "Pressure Low Set Point".
- **Pressure Maximum to Alarm:** Is the maximum pressure (Bar) allowable in the equipment.
- **Pressure Maximum to work:** is the maximum pressure (Bar) to alarm set closer to the working pressure to protect the system and alarm of process changes.
- **Pressure Maintain:** This is the pressure (Bar) set when using the pressure maintain feature. Pressure maintain is enabled when a value is inserted, if you do not wish to use this feature, leave the value at zero. The pressure maintain feature allows the pump to turn at a slow rate when the gear pump is idle, the function is to hold the pressure in the line as close to the normal extrusion pressure and therefore in normal working practice it will be sent 5-10 Bar lower than the normal operating extrusion pressure.

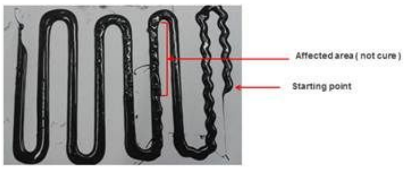
- **Real Pressure (Bar):** Displays the actual pressure being read between the gear pump and applicator head at that time.
- **Read Point:** Displays the "points" value received from the sensor between the gear pump and applicator head at that time.

- **Sensor Pressure Pump B:** Title indicating that all below values are related to the pressure sensor situated on the exit of the B Component gear pump.
- **Pressure Max to set:** Upper pressure value expressed in Bar to calibrate the pressure sensor.
- **Set Min Pressure:** Lower pressure value expressed in Bar to calibrate the pressure sensor.
- **Point Maximum Set :** Upper value expressed in points that represents the analogue signal received from the pressure sensor relating to the pressure input in the parameter "Pressure High Set Point".
- **Point Minimum Set :** Lower value expressed in points that represents the analogue signal received from the pressure sensor relating to the pressure input in the parameter "Pressure Low Set Point".
- **Pressure Maximum to Alarm:** Is the maximum pressure (Bar) allowable in the equipment.
- **Pressure Maximum to work:** is the maximum pressure (Bar) to alarm set closer to the working pressure to protect the system and alarm of process changes.
- **Pressure Maintain:** This is the pressure (Bar) set when using the pressure maintain feature. Pressure maintain is enabled when a value is inserted, if you do not wish to use this feature, leave the value at zero. The pressure maintain feature allows the pump to turn at a slow rate when the gear pump is idle, the function is to hold the pressure in the line as close to the normal extrusion pressure and therefore in normal working practice it will be sent 5-10 Bar lower than the normal operating extrusion pressure.
- **Real Pressure (Bar):** Displays the actual pressure being read between the gear pump and applicator head at that time.
- **Read Point:** Displays the "points" value received from the sensor between the gear pump and applicator head at that time.

8. Fault Finding

Fault	Checks	Action
High Pressure Alarm	Pressure Value	Check pressure sensor value is being read, When sensor disconnected it reads approx. -61.3. Replace sensor and re check.
	When was the glue mixer was last changed	If the mixer was not changed for more than 30-45 minutes, replace and re check.
	Obstructions in adhesive flow	Check for obstruction in adhesive flow from the gear pump the mixer exit such as cured adhesive.
	Has the glue pump speed been changed	If the pump speed is increased the line pressure will increase, adjust the pressure settings to suit the new speed (Pressure maintain 5Bar less than the running pressure / Max Pressure 15 Bar higher than running pressure).
Pump Shaft Broken	Pressure Value	If the value is a negative value and “Pressure maintain” is enabled the pump will continue to turn until the shaft breaks. Check and replace sensor.
		Check pressure history, if its been increasing then material has been curing in part of the line blocking the flow, identify restriction and clear.
	Obstruction in pump	Solid particle in the pump will block gear causing the shaft to break. Check for particles when replacing the shaft.
	3 Way Valve	Check the 3 way valve position after the gear pump and ensure the direction of flow is correct.
Emergency Thermic Protection alarm on PLC	Circuit breaker for motor has been tripped	Request a qualified electrician check the circuit breaker supply to the motors in the control panel.
Emergency Air pressure control alarm on PLC	Air pressure Supply is not sufficient	Check air pressure supply to the machine, Minimum 5 Bar.
		Check the air stop valve is in the correct position on the air feed.
		Check the emergency button is not pressed and the panel has been reset “Start-up”.
		Check the digital input signal from the pressure present sensor in Test I/O

Fault	Checks	Action

Part Failed Leak test	Snake Test  Please take note that the glue is not cured consistently at the particular area	If there is a an area of uncured or slow curing adhesive part way into the snake test but the rest of the material is curing normally then the problem is relating to when the gun opens to start the extrusion. Check both sides of the gun are open and closing simultaneously. Check the pressure for A and B is as normal, a higher pressure difference may cause one part to restrict the initial flow of the second part when the gun opens and whilst the flow equalises, identify the cause and repair, typical cause is cured adhesive in the gun exit block.
	Adhesive flow from the mixer nozzle	Is the flow less smooth than normal, change the mixer nozzle and retest. Higher than normal ambient temperatures can cause the pot-life of the adhesive to be shortened.
	Check the adhesive extrusion without the lens	Is the extrusion complete inside the housing channel. Check the run signal to the Dynamix from the robot is constant.
		If the flow is intermittent and air popping sounds can be heard air has entered the line, checking all joints and ensure that during drum changes the material is purged to prevent air entrapment.
	Adhesive flow without the mixer nozzle and with the flow separator attached	If the flow is visibly reducing and the pressure is reducing check the supply to the dosing pumps from the transfer pumps and that the glue feed supply valves mounted on the dosing pump manifold are opening when the run signal is given.
	Check the surface tension of the housing	If the surface tension is lower than required check the Plasma function.
	Mix ratio check	If the mix ration is not correct check values in the PLC, if these are correct check for a cause of flow change from the transfer pumps to the glue head. Isolate the parts starting from the glue drums and identify where the flow restriction is originating. Check the age of the gear pump as it may be consumed and not maintaining the correct flow.

Fault	Checks	Action
Pump not turning	Request a technician check the screen on the Sanyo driver for alarms.	If an alarm is present refer to the Sanyo manual, if no alarm is present check that the display icon moves in a circular motion when the start signal is given. AL85 suggests a problem with the wiring connections or with the driver. Check the cables if its still remains replace the driver.
	Is the motor shaft turning but the pump not.	Check the glue speed is not at zero.
		Check the pump shaft is not broken, if yes refer to “Pump shaft
		Check the pump shaft is not broken, if yes refer to “Pump shaft broken” section of fault finding.

9. Recording data

It is recommended to log data throughout production to assist with root cause analysis should there be any issues during production. Without basic data it is difficult to identify the cause of any variation in leak test failures throughout production.

An example of a potential data logging sheet is as follows:

Date	Time	Operating Pressure			Mix Ratio			Drum Change		Other
		Pressure A	Pressure B	Pressure stability (stable/Not)	Qutantity A Component	Quantity B Component	Ratio A/B	A1 / A2	B1 / B2	

Form Code : DM-DYM-REC001

Operating pressure: Record the pressure in the line during the extrusion.
Pressure stability: Monitor the pressure thoughout the extrusion, indicate if the pressure is stable throughout the extrusion. Inidicate any abnomalities in the "Other" column.
Mix ratio: Using the adapter provided by Diemme Fin to divide the two components, in the "Balance" page on the PLC purge the two components into a cup based with the quantity set at the usual extrusion quantity and the time set for 60 seconds.
Drum Change: Indicate in the box the drum changed and in the "Time/Date" columns the time and date of the change.
Other: Make a note of any points/component failures related to the glue application equipment.

FINDIEMME
ADHESIVE TECHNOLOGIES

10. Heated hoses

The heated hoses have an independent heater module attached to the hose, therefore they require simply a 220V supply. The hose temperature is self-regulated.

You can see the hose is heating by the red LED on the side of the small local electrical box on the hose.

To change the temperature that the hose maintains it is necessary for a qualified electrician to isolate the power supply and open the small electrical box on the hose. Inside there are a selector switches, the position of these switches dictates the temperature maintained as follows:

SW 6	SW 5	SW 4	SW 3	SW 2	SW 1	set-point °C
on	on	on	on	on	on	25
on	on	on	on	on	off	28
on	on	on	on	off	on	31
on	on	on	on	off	off	34
on	on	on	off	on	on	37
on	on	on	off	on	off	40
on	on	on	off	off	on	43
on	on	on	off	off	off	46
on	on	off	on	on	on	49
on	on	off	on	on	off	52
on	on	off	on	off	on	55
on	on	off	on	off	off	58
on	on	off	off	on	on	61
on	on	off	off	on	off	64
on	on	off	off	off	on	67
on	on	off	off	off	off	70
on	off	on	on	on	on	73
on	off	on	on	on	off	76
on	off	on	on	off	on	79
on	off	on	on	off	off	82
on	off	on	off	on	on	85
on	off	on	off	on	off	88
on	off	on	off	off	on	91
on	off	on	off	off	off	94
on	off	off	on	on	on	97
on	off	off	on	on	off	100
on	off	off	on	off	on	103
on	off	off	on	off	off	106
on	off	off	off	on	on	109
on	off	off	off	on	off	112
on	off	off	off	off	on	115
on	off	off	off	off	off	118

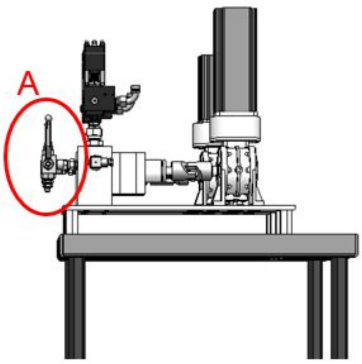
b) temperatura attualmente programmata indicata da una croce

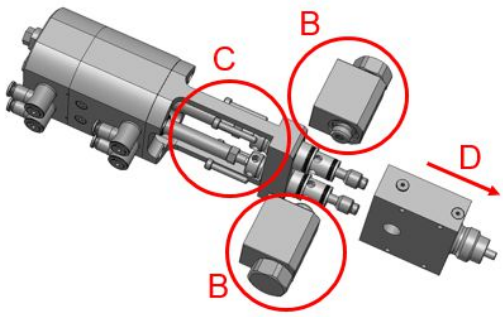
sw 6	sw 5	sw 4	sw 3	sw 2	sw 1	set-point °C
off	on	on	on	on	on	121
off	on	on	on	on	off	124
off	on	on	on	off	on	127
off	on	on	on	off	off	130
off	on	on	off	on	on	133
off	on	on	off	on	off	136
off	on	on	off	off	on	139
off	on	on	off	off	off	142
off	on	off	on	on	on	145
off	on	off	on	on	off	148
off	on	off	on	off	on	151
off	on	off	on	off	off	154
off	on	off	off	on	on	157
off	on	off	off	on	off	160
off	on	off	off	off	on	163
off	on	off	off	off	off	166
off	off	on	on	on	on	169
off	off	on	on	on	off	172
off	off	on	on	off	on	175
off	off	on	on	off	off	178
off	off	on	off	on	on	181
off	off	on	off	on	off	184
off	off	on	off	off	on	187
off	off	on	off	off	off	190
off	off	off	on	on	on	193
off	off	off	on	on	off	196
off	off	off	on	off	on	199
off	off	off	on	off	off	202
off	off	off	off	on	on	205
off	off	off	off	on	off	208
off	off	off	off	off	on	211
off	off	off	off	off	off	214

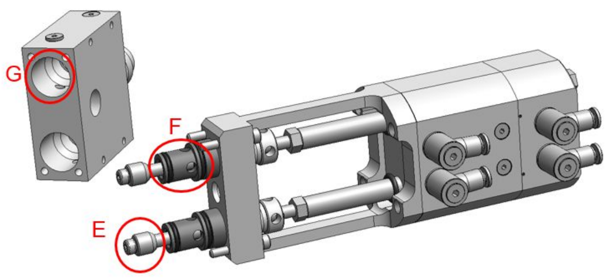
11 Other

11.1 Cleaning the Ezi Mix Gun if contaminated.

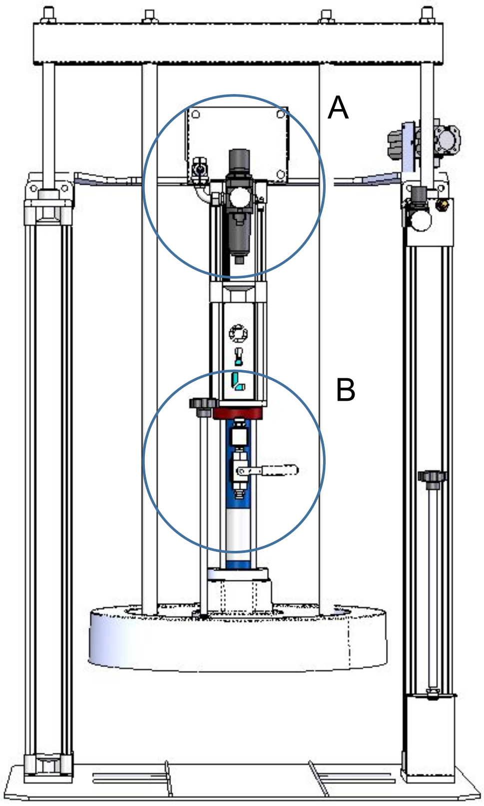
1. Exit Automatic to disable pressure maintain.
2. Remove the pressure from the line by moving the 3 way valve after the gear pump to the purge position (A).
3. Disconnect the glue feeds to the gun (B).
4. Remove the 4 bolts that hold the glue gun block to the gun housing (C).
5. Remove the glue gun block (D).
6. Clean any solid material from the pistons (E) / seats (F) & gun block (G) that may prevent the gun from being able to close completely.
7. 6. Make sure the seats / pistons and blocks are clear for both components. Take care with the piston as this is a hard plastic, do not use sharp objects to clean.
8. 7. Check the piston and seat for damage, replace if necessary.
9. 8. Reassemble and test.
10. 9. Purge the line to make sure there is no air in the system before restarting production.







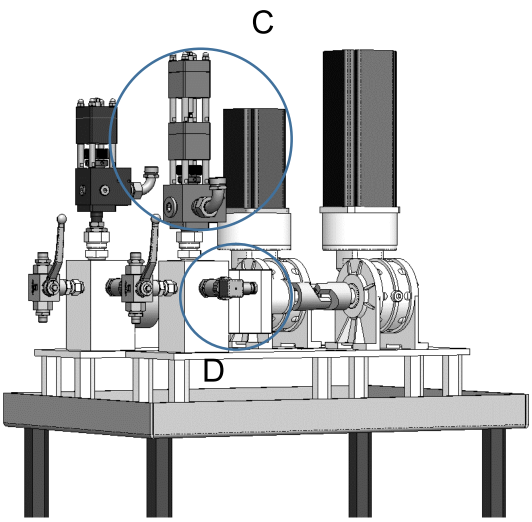
11.2 A Component Transfer Pump Line pressure control



- The Pressure regulator "A", controls the Air pressure supply to the Piston Transfer Pump.
- Due to Air cylinder " B" to Piston Ram ratio the air pressure in, creates a fluid pressure out with a ratio of 1:25 Bar. Therefore 3 Bar Air pressure = 75 Bar Fluid Pressure at the exit "B".
- The 2 way valve "C", has two functions. A) to hold the line pressure ahead of the valve. B) to prevent the high fluid pressure creating a force on the gear pump while the pump is not enabled (the gaps between extrusions).
- If the Fluid pressure from "B" is higher than the force to open the two way valve "C" then the valve will not open.
- The extrusion pressure is measured after the two way valve and after the gear pump "D".

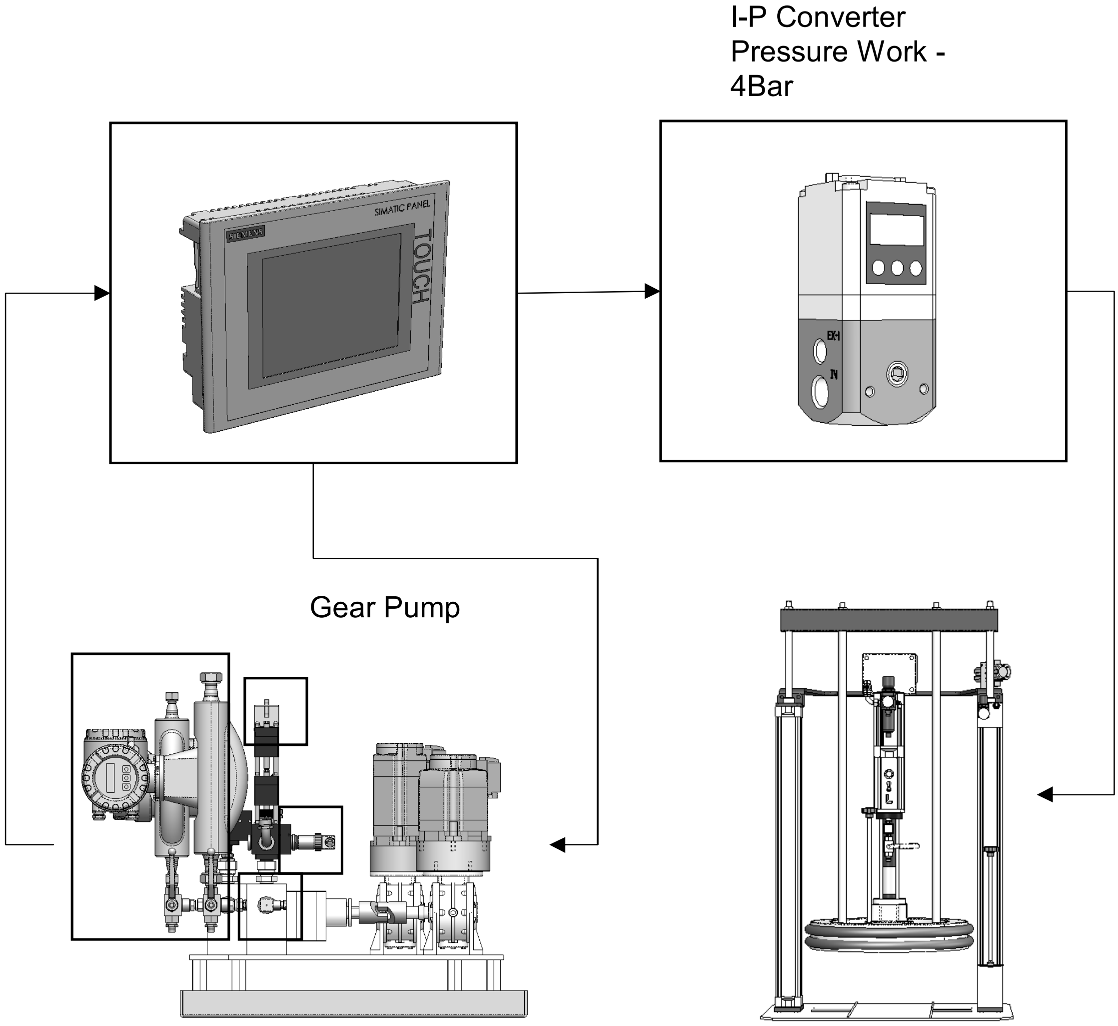
If the Minimum pressure alarm occurs, check the following:

1. The Air pressure on the transfer pump (A) is not too high (for normal applications 3 Bar is sufficient).
2. Check the flow from the purge valve (B) to be sure that the flow is sufficient based on the Air pressure set, this should be set at the minimum possible to achieve sufficient guaranteed flow. Set the pressure, purge from the local purge valve and observe the flow and the stroke of the piston.
3. Check that the Valve (C) can open/close repeatedly with the fluid pressure set and the site pressure available. If not add an air booster to the system to increase the air pressure to the valve.



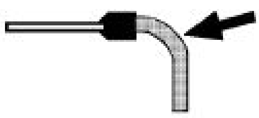
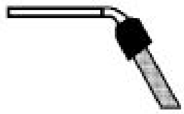
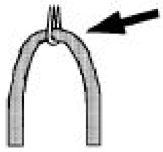
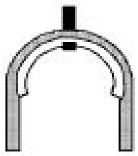
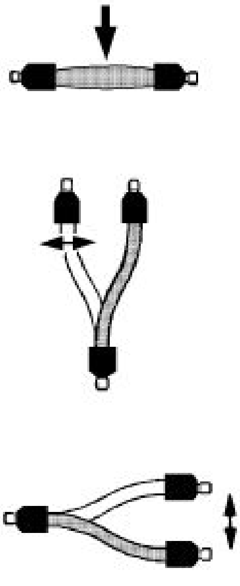
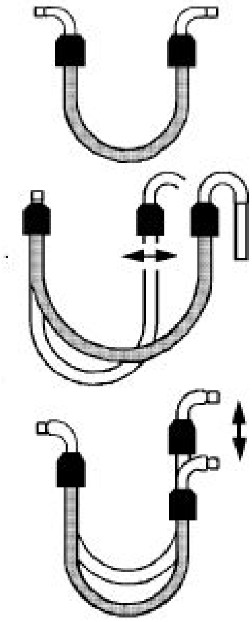
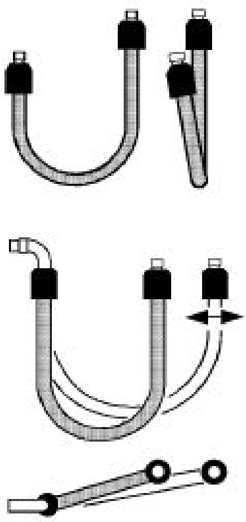
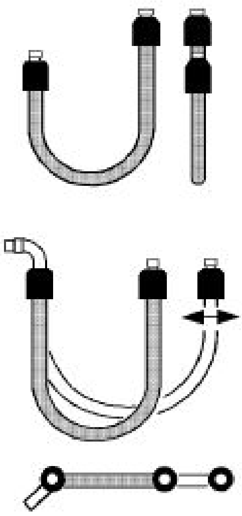
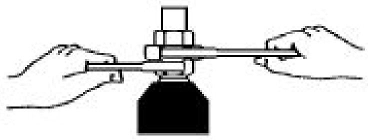
Note: If the minimum pressure alarm occurs because the valve (C) cannot open due to the fluid pressure being too low and you increase the air pressure (A) to the transfer pump, the problem will become worse not better.

11.3 A Component Control Flow - Basic



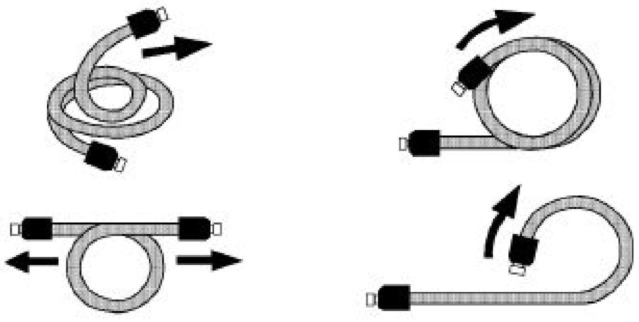
Data feedback to the PLC (Gear Pump Module):
A. Line Pressure Pre Gear Pump (Bar)
B. Line Pressure Post Gear Pump (Bar)
C. Material data (Flow/Temperature/Density)
D. Valve Position sensor indicator
Data feedback to the PLC (Transfer Pump):
A. Low Level drum

11.4 Mounting Hoses guide lines (Heated hose)

Wrong	Right	
		If a hose is connected at 90°, the effect of its weight will cause a harmful bend.
		Continually moving hoses are subject to bending. It is advisable to mount a bend restrictor or a suitably sized pulley as a support.
		Compressing a hose lengthwise, mounting it incorrectly or movement during use can cause variations in pressure continuity. Advice: mount the hoses with a 90° connector and a wide bend.
		Incorrect mounting can cause hose torsion, particularly at the moment of the installation. It is advisable to make certain that the hose's axes are parallel and that the movement is on one plane. It is also advisable to use two keys when tightening the hose fitted to the system. 

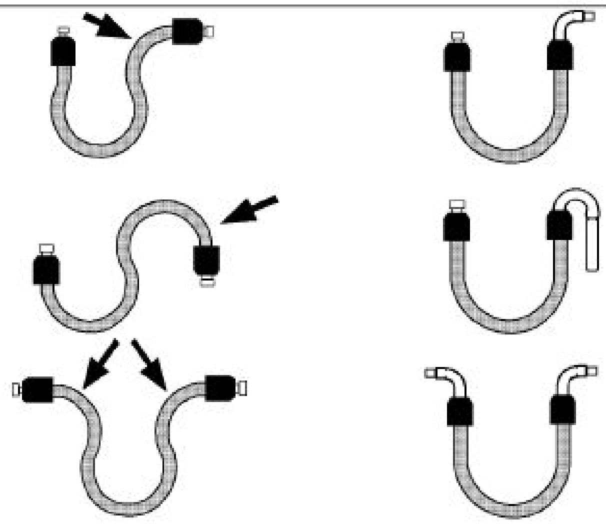
11.4 Mounting Hoses guide lines (Heated hose)

Wrong **Right**

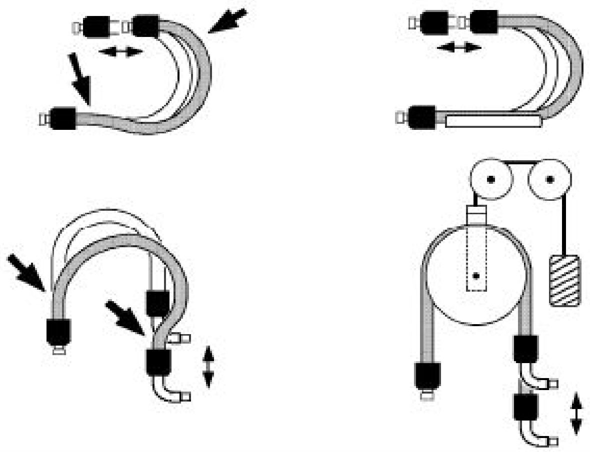


Pulling one end of a coiled hose can cause a bend in the hose and exceed the minimum bend radius.
Hint: unwind the hose without pulling.

When connected, hoses which are too short can form bends which are too sharp near the fittings.
Hint: always add 50mm. extra



Excessively sharp bends near the ends always damage the hose. Electrically heated hoses' most delicate points are always near the ends. It is advisable to take great care at the above mentioned point.



If a hose is mounted incorrectly, its weight can cause it to bend. Help it with a U-shaped piece of sheet metal or pulleys and counter-weights.